



Sustainability-Oriented Microgrid Selection for Rural Clinics Using Linguistic q-Rung Orthopair Fuzzy Hypersoft Technique for Order Preference by Similarity to Ideal Solution

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Abstract: Reliable and sustainable electricity supply remains a critical challenge for rural healthcare facilities, where energy planning involves competing technical, economic, environmental, social, and resilience considerations under substantial uncertainty. Conventional multi-criteria decision models may have difficulty representing linguistic assessments, hesitation, and the hierarchical structure of subdivided evaluation attributes. This study develops a sustainability-oriented decision-support framework for resilient microgrid selection by integrating linguistic q-rung orthopair fuzzy hypersoft sets (Lq-ROFHSs) with an extended technique for order preference by similarity to ideal solution (TOPSIS) model. Correlation and weighted correlation coefficients for Lq-ROFHSs were formulated and their mathematical properties were examined. These measures were then incorporated into a multi-attribute group decision-making (MAGDM) procedure in which expert assessments, attribute weights, and subdivided criteria were represented within a unified linguistic fuzzy structure. The framework was applied to the selection of microgrid alternatives for rural clinics across cost efficiency, reliability, environmental benefit, and scalability-related criteria. The analysis produced a clear preference ordering among the evaluated alternatives, while comparison with existing decision-making approaches showed consistent ranking behaviour and demonstrated the applicability of the proposed formulation to structured decisions under linguistic uncertainty. The results indicate that combining hypersoft attribute representation with correlation-based TOPSIS provides a transparent framework for sustainability-oriented evaluation involving complex and imprecise information. The proposed approach offers a reproducible decision-analytic basis for microgrid planning and can be extended to other sustainability and energy-system decisions involving multiple stakeholders, competing criteria, and uncertain assessments.

Keywords: Sustainability decision analytics; Resilient microgrid selection; Linguistic q-rung orthopair fuzzy hypersoft sets; Multi-attribute group decision-making; Technique for order preference by similarity to ideal solution; Uncertainty modelling

1 Introduction

Smart-energy systems, distributed generation, and decentralized energy infrastructure are rapidly evolving, creating new opportunities to improve the reliability and sustainability of electricity supply, particularly in remote and rural communities. Microgrids have emerged as an important option for integrating renewable energy resources, energy storage systems, and distributed generation while strengthening local energy resilience. Their role is particularly important for rural healthcare facilities, where clinical operations, communication systems, vaccine storage, and essential medical equipment depend on a reliable electricity supply. Selecting an appropriate microgrid configuration, however, is a complex decision problem involving multiple technical, economic, environmental, social, and resilience-related criteria. These criteria may conflict with one another, while heterogeneous stakeholder preferences, incomplete information, and linguistic assessments introduce additional uncertainty into the evaluation process.

These characteristics create a need for structured decision-support frameworks capable of processing complex and

uncertain information and producing transparent and reliable preference rankings. Multi-attribute group decision-making (MAGDM) provides a suitable basis for problems in which multiple alternatives are evaluated against several attributes by a group of decision makers. In practical sustainability and energy-planning problems, however, expert assessments are often expressed linguistically, and individual attributes may need to be decomposed into several related sub-attributes. Conventional fuzzy models may not adequately preserve this level of information granularity. More flexible mathematical structures are therefore required to accommodate linguistic uncertainty, hesitation, and the multidimensional nature of subdivided decision attributes.

In decision-making theory, MAGDM provides a systematic framework for combining quantitative and qualitative information when alternatives must be assessed by a group of decision-making experts $\mathcal{DM}_e$. MAGDM is particularly relevant when multiple, often conflicting, factors must be considered simultaneously and the participating $\mathcal{DM}_e$ hold different preferences or judgements. Incorporating diverse expertise within a common evaluation structure can provide more comprehensive rankings, reduce the influence of individual bias, and make the decision process easier to interpret and justify. In many practical settings, however, the available information is vague or imprecise, making it difficult for $\mathcal{DM}_e$ to provide exact numerical evaluations. Recognizing this problem, Zadeh [1] introduced fuzzy set (FS) theory to represent imprecise information quantitatively. A fuzzy set A in a universe of discourse X $\mu_A : X \rightarrow [0, 1]$, is characterized by a membership function that assigns each element $x \in X$ a degree of membership $\mu_A(x)$. Although FS theory provides an important foundation for uncertainty modelling, its representation relies entirely on a single membership degree and may therefore be insufficient when decision information contains additional forms of hesitation or ambiguity. Fuzzy-set-based methods have subsequently been applied to a broad range of practical problems. Zadeh [2] introduced linguistic quantifiers for fuzzy reasoning. Muneeza Ihsan and Abdullah [3] and Noor et al. [4] investigated fuzzy methods for multi-criteria group decision-making, including applications involving COVID-19 testing and probabilistic hesitant fuzzy operators. Zadeh et al. [5] and Farman et al. [6] examined fuzzy clustering and decision-making under T-spherical fuzzy environments. Saqlain et al. [7] contributed to the development of methods for the rural health services. Alhamzi et al. [8] proposed a Dombi-based aggregation operators, while Wu et al. [9] proposed heterogeneous linguistic expressions for group decision-making. Saqlain et al. [10] introduced linguistic neutrosophic method for medical diagnosis and treatment, in another study Saqlain et al. [11] proposed a method for smart farming decisions under weather uncertainty. While Saqlain et al. [12] developed a multi-criteria decision-making (MCDM) framework for complex decision environments. Together, these studies illustrate the broad development and applicability of fuzzy logic across decision-making and computational environments.

Atanassov [13] addressed an important limitation of conventional fuzzy sets by introducing intuitionistic fuzzy sets (IFSs), which incorporate both membership μ and non-membership ν , subject to the condition $\mu + \nu \leq 1$. This representation provides additional information about uncertainty, but it cannot accommodate assessments for which $\mu + \nu > 1$. Yager [14] introduced Pythagorean fuzzy sets (PFSs) to relax this restriction by requiring $\mu^2 + \nu^2 \leq 1$, thereby providing $\mathcal{DM}_e$ with a broader admissible range for expressing their evaluations. Nevertheless, some assessments may still satisfy $\mu^2 + \nu^2 > 1$. Yager [15] therefore proposed q-rung orthopair fuzzy sets (q-ROFSs), for which $\mu^q + \nu^q \leq 1$ with $q \geq 1$, further expanding the feasible space for representing uncertain decision information. These developments, together with different hybrid fuzzy structures and aggregation operators, have supported the construction of increasingly flexible MAGDM methods. Alcantud [16] investigated intuitionistic fuzzy aggregation operators defined by weighted geometric means for MAGDM under uncertainty. Liu et al. [17] subsequently developed intuitionistic fuzzy partitioned Maclaurin symmetric mean operators for group decision-making. Garg and Chen [18] proposed neutrality aggregation operators for q-ROFSs, providing additional flexibility in decision analysis. Pathak et al. [19] developed a multi-criteria decision-making framework incorporating Einstein power operators and interval-valued q-ROFSs. Kumar and Kumar [20] proposed an intuitionistic fuzzy similarity measure for decision-making and pattern-recognition problems, while Kumar and Chen [21] developed an intuitionistic fuzzy weighted averaging aggregation operator to incorporate linguistic information into the evaluation process. Despite the increasing flexibility of IFSs, PFSs, and q-ROFSs, decision-makers ($\mathcal{DM}_e$) may still find it difficult to express some assessment information numerically. This is particularly relevant when evaluations are naturally qualitative, as in descriptions of weather conditions or subjective expert judgements. Zadeh [22] introduced the concept of a linguistic variable (LV) to represent such information, providing an important basis for decision models involving qualitative assessments. More recent developments in group decision-making have consequently placed increasing emphasis on linguistic models for complex multi-expert environments.

Herrera and Martínez [23] introduced a linguistic 2-tuple model for handling multigranular hierarchical linguistic contexts, allowing linguistic information expressed at different levels of granularity to be managed in multi-expert decision-making. Xu [24] developed linguistic aggregation operators for group decision-making with linguistic preference relations, facilitating the aggregation of heterogeneous linguistic assessments. Saha et al. [25] proposed a dual probabilistic linguistic consensus-reaching method for group decision-making. Akram et al. [26] further developed an extended Multi-Objective Optimization by Ratio Analysis plus the Full Multiplicative Form

(MULTIMOORA) method based on 2-tuple linguistic Pythagorean fuzzy sets for multi-attribute group decision problems. These studies progressively increased the ability of linguistic decision models to accommodate heterogeneous expert assessments and complex preference information.

Chen et al. [27] combined LVs with intuitionistic fuzzy numbers (IFNs) to develop linguistic intuitionistic fuzzy sets (LIFSs), providing a more flexible representation of qualitative assessments. Garg [28] subsequently extended this concept to linguistic Pythagorean fuzzy sets (LPFSs) by integrating LVs with PFSs. Han et al. [29] and Lin et al. [30] developed technique for order preference by similarity to ideal solution (TOPSIS) methods based on entropy, distance measures, and correlation coefficients for decision problems involving LPFSs. Liu and Liu [31] further extended this line of research by introducing Lq-ROFNs, in which the membership and non-membership information is expressed using linguistic variables. Building on this formulation, a number of MAGDM approaches under the Lq-ROFN environment have been developed using power Bonferroni aggregation operators, power Muirhead mean operators, similarity measures, and Einstein aggregation operators.

Research on linguistic q-rung orthopair fuzzy information has subsequently expanded toward more sophisticated group decision models under uncertainty. Akram et al. [32] developed a group decision-making framework based on linguistic q-rung orthopair fuzzy Einstein models. Liu et al. [33] proposed linguistic q-rung orthopair fuzzy generalized point weighted aggregation operators for group decision analysis. Neelam et al. [34] introduced Yager prioritized weighted geometric aggregation operator for Lq-ROFs to quantify uncertainty and information within this environment. Li and Zhang [35] proposed a cognitively inspired group decision-making method using linguistic q-rung orthopair fuzzy preference relations. Liu and Liu [36] developed a multiple-attribute group decision-making method based on linguistic q-rung orthopair fuzzy power Muirhead mean operators with entropy weights to represent decision-makers' preferences. Peng et al. [37] further investigated projection-based similarity measures for linguistic q-rung orthopair fuzzy multi-criteria group decision-making. Together, these studies demonstrate the flexibility of linguistic q-rung orthopair fuzzy structures for representing uncertain and linguistically expressed evaluations across a broad range of decision problems.

MAGDM approaches involving Lq-ROFNs, weighted aggregation operators, and preference orders of alternatives have also been investigated in sustainability-oriented applications [38]. A further development relevant to complex attribute structures is the q-rung orthopair fuzzy soft set (q-ROFSS), for which Hussain et al. [39] introduced associated aggregation operators. Smarandache [40] subsequently introduced the hypersoft set (HSS), extending the soft-set framework by allowing parameters to be decomposed into distinct sub-attributes. This characteristic makes HSS particularly suitable for decision problems in which individual evaluation criteria contain several related dimensions. Khan et al. [41] developed q-ROFHSs and established their basic operations. Aggregation operators and weighted aggregation operators with practical applications were subsequently investigated by Zulqarnain et al. [42–44] and Ying et al. [45]. Gurmani et al. [46] further developed correlation coefficients for q-ROFHSs. More recent studies have extended q-rung orthopair fuzzy models to a range of decision settings. Mandal and Seikh [47] proposed an Evaluation based on Distance from Average Solution (EDAS)-based method for selection problems involving q-rung orthopair fuzzy information. Seikh and Mandal [48] applied q-rung orthopair fuzzy Frank aggregation operators to multiple-attribute decision-making. They subsequently developed a MAGDM approach for electric vehicle charging-station site selection [49]. Seikh and Mandal [50] also investigated q-rung orthopair fuzzy Archimedean aggregation operators for software operating-unit selection and later examined a 3,4-quasirung fuzzy framework for more general multiple-attribute decision problems [51].

Recent developments in decision analysis have increasingly combined hypersoft structures with other computational and uncertainty-modelling techniques to address complex practical problems. Saqlain [52] developed a sustainable hydrogen-production framework by combining ViseKriterijumska Optimizacija I Kompromisno Resenje (VIKOR; Multi-Criteria Optimization and Compromise Solution) with intuitionistic hypersoft sets, demonstrating the relevance of hypersoft modelling to sustainability-oriented energy decisions under uncertainty. Hamid and Abid [53] developed a decision-support system for mobile-phone selection using fuzzy hypersoft sets and machine learning to manage multiple criteria and uncertain information. These studies, together with the preceding developments in linguistic q-rung orthopair fuzzy modelling, illustrate a broader movement toward decision frameworks capable of preserving increasingly rich forms of uncertain, linguistic, and attribute-level information. Such capabilities are particularly relevant to sustainability-oriented energy decisions, where alternatives must be compared across several competing dimensions while expert assessments are often imprecise and individual criteria may require further subdivision.

1.1 Motivation and Research Problems

Considerable progress has been made in MAGDM through fuzzy sets, intuitionistic fuzzy sets, q-ROFSSs, linguistic extensions, and related aggregation mechanisms. An important methodological challenge nevertheless remains when decision attributes must be further subdivided and evaluations are simultaneously obtained from multiple experts. In such settings, the decision model must preserve not only uncertainty and linguistic hesitation but also the internal structure of the subdivided attributes. Existing linguistic q-rung orthopair fuzzy approaches

provide flexible representations of uncertain expert assessments, while hypersoft structures allow parameters to be decomposed into multiple sub-attributes. However, the integration of these capabilities within correlation-based MAGDM remains comparatively underdeveloped. In particular, correlation and weighted correlation measures are needed to characterize relationships between Lq-ROFHS information and to support structured preference ordering within a transparent decision procedure. This issue is particularly relevant to sustainability-oriented problems such as resilient microgrid selection, where technical, economic, environmental, social, and resilience considerations may contain multiple sub-attributes and depend on linguistic assessments from several decision makers.

Accordingly, this study addresses the following research questions:

1. How can correlation and weighted correlation coefficients be defined and mathematically examined for linguistic q-rung orthopair fuzzy hypersoft sets (Lq-ROFHSs)?
2. What fundamental properties should the proposed correlation coefficients satisfy to provide a consistent basis for MAGDM under Lq-ROFHS information?
3. How can the proposed weighted correlation coefficient be integrated with TOPSIS to construct and implement a MAGDM procedure in an Lq-ROFHS environment?
4. How does the proposed MAGDM framework compare with existing methods in deriving rankings and preference orders for complex decision problems?
5. How can the proposed framework be applied to a sustainability-oriented decision problem involving resilient microgrid selection under multidimensional and linguistically expressed expert assessments?

To address these questions, this study develops correlation and weighted correlation coefficients for Lq-ROFHSs and examines their mathematical properties. These measures are incorporated into an extended TOPSIS procedure to construct a correlation-based MAGDM framework capable of preserving linguistic uncertainty, hesitation, and subdivided attribute information. The resulting framework is then applied to resilient microgrid selection for rural clinics, where alternatives are evaluated across technical, economic, environmental, social, and resilience-related dimensions. In this way, the methodological development is linked to a structured sustainability decision problem rather than being considered solely as an abstract extension of fuzzy-set theory.

1.2 Contribution and Layout of the Study

The principal contribution of this study is the development of correlation and weighted correlation coefficients for Lq-ROFHSs and their integration into a TOPSIS-based MAGDM framework. The proposed measures provide a means of characterizing relationships between Lq-ROFHS information while retaining the linguistic membership and non-membership assessments, uncertainty, and subdivided attribute structure represented by the hypersoft environment. Their mathematical properties are examined to establish the analytical basis of the proposed formulation. Incorporating these correlation measures into TOPSIS subsequently provides a structured mechanism for differentiating preference orders among competing alternatives under multi-expert and multi-sub-attribute information.

The proposed approach can also serve as the analytical basis of an intelligent decision-support system (IDSS) that transforms linguistic expert judgements into systematic and interpretable alternative rankings. The decision process begins with the identification of alternatives and evaluation criteria, followed by the elicitation of linguistic assessments from the participating decision makers. These assessments are represented within the Lq-ROFHS framework, allowing uncertainty, hesitation, and further subdivisions of individual attributes to be retained. The proposed correlation and weighted correlation coefficients are then used to characterize relationships within the evaluation information. The extended TOPSIS procedure subsequently calculates the relative closeness of each alternative to the ideal solution, providing the basis for preference ordering and final alternative selection.

From an application perspective, the framework provides a systematic means of incorporating heterogeneous expert knowledge when precise numerical information is unavailable or difficult to obtain. Its linguistic representation allows experts to express judgements in qualitative terms, while the q-rung orthopair fuzzy and hypersoft structures preserve uncertainty and attribute-level detail throughout the decision process. In the present study, these characteristics are examined through resilient microgrid selection for rural clinics, where sustainability-oriented decision-making requires the simultaneous consideration of technical, economic, environmental, social, and resilience criteria. The resulting preference rankings, together with comparative and sensitivity analyses, provide a basis for examining the behaviour and stability of the proposed decision procedure. The framework is also relevant to other industrial, sustainability, and smart-energy decisions characterized by multiple stakeholders, competing criteria, subdivided attributes, and uncertain assessments.

The remainder of the paper is organized as follows. Section 2 presents the mathematical preliminaries required for the study, including linguistic quantifiers, HSs, and q-ROFHSs. Section 3 develops the correlation and weighted correlation coefficients within the Lq-ROFHS environment and examines their mathematical properties. Section 4 integrates the proposed measures into an extended TOPSIS-based MAGDM procedure. Section 5 presents the application of the framework to resilient microgrid selection for a rural clinic and illustrates its numerical

implementation. Section 6 provides comparative and sensitivity analyses and discusses the theoretical and managerial implications of the results. Finally, Section 7 summarizes the main findings, discusses the limitations of the proposed framework, and outlines directions for future research.

2 Preliminaries

This section introduces the fundamental concepts and mathematical notation required for the development of the proposed framework. The definitions presented below provide the theoretical basis for the linguistic q-rung orthopair fuzzy hypersoft structure and the subsequent correlation-based decision model.

Definition 1. [2] Linguistic quantifiers are expressions used in natural language to describe quantities in an imprecise or qualitative manner, such as low, medium, and high. Originally introduced by Zadeh, these quantifiers provide a linguistic representation of qualitative assessments and are summarized in Table 1.

Table 1. Linguistic quantifiers

Quantifiers						
Low		Medium			High	
None	Very-Low	Low	Medium	High	Very-High	Perfect

Let $K = \{\kappa^1, \kappa^2, \kappa^3, \dots, \kappa^t\}$ where $t = 2n + 1 : n \geq 1$ and $n \in \mathbb{R}^+$. The set (K) is assumed to be finite and strictly increasing.

Definition 2. [34] Consider $\mathfrak{H}, \mathcal{U}$, and $\mathcal{P}(\mathcal{U})$ be the set of attributes, universe of discourse, and power set of universes, respectively. Let $\mathfrak{H} = \{\mathfrak{H}^1, \mathfrak{H}^2, \mathfrak{H}^3, \dots, \mathfrak{H}^p\}$ ($p \geq 1$), then assume q-ROHS be a collection of all q-rung orthopair subsets over $\mathcal{U}$. Then the pair $(\mathcal{W}, \mathfrak{H}^1 \times \mathfrak{H}^2 \times \mathfrak{H}^3 \times \dots \times \mathfrak{H}^p) = (\mathcal{W}, \mathfrak{H})$ is known as q-Rung Orthopair hypersoft set (q-ROHS). Represented by Eq. (2) mathematically:

$$\mathcal{W} : \mathfrak{H}^1 \times \mathfrak{H}^2 \times \mathfrak{H}^3 \times \dots \times \mathfrak{H}^p = \mathfrak{H} \rightarrow \mathcal{P}(\mathcal{U})_q - \text{ROHS} \quad (1)$$

and defined as $(\mathcal{W}, \mathfrak{H}) = \{(\mathcal{T}_{\mathcal{W}}(e), \mathcal{J}_{\mathcal{W}}(e)) : e \in \mathfrak{H}\}$, where $(\mathcal{T}_{\mathcal{W}}(e), \mathcal{J}_{\mathcal{W}}(e))$ represent membership and non-membership of attributes, respectively, such that

$$0 \leq (\mathcal{T}_{\mathcal{W}}(e))^q + (\mathcal{J}_{\mathcal{W}}(e))^q \leq 1 \quad (2)$$

Definition 3. [40] The pair $(\mathcal{W}, \mathfrak{H})$ is called an HSS over $\mathcal{U}$, where $\mathfrak{H}$ is the cartesian product of n disjoint sets $\mathfrak{H}_1, \mathfrak{H}_2, \mathfrak{H}_3, \dots, \mathfrak{H}_p$ having attribute values of p distinct attributes $\mathfrak{H}^1, \mathfrak{H}^2, \mathfrak{H}^3, \dots, \mathfrak{H}^p$, respectively. Represented by Eq. (1) mathematically:

$$\mathcal{W} : \mathfrak{H} \rightarrow \mathcal{P}(\mathcal{U}) \quad (3)$$

3 Proposed Correlation Measures

This section is further divided into two subsections. Subsection 1 presents the definition of Lq-ROHSs and Lq-ROFHSs. Subsection 2 features the informational energies and correlation coefficient for Lq-ROFHSs. Consider $\mathcal{E}(\mathcal{R})_{[0,t]}$ be the set of all Lq-ROFHSs over the universal set $\chi = \{x^1, x^2, \dots, x^n\}$, where μ and ν of each element $x^i \in \mathcal{R}$ belongs to the linguistic quantifiers $\ell_{[0,t]}$.

3.1 Correlation Measures

This subsection presents the definitions along with solved examples of Lq-ROHSs and Lq-ROFHSs.

Definition 4. Consider χ, a_n , and $\mathcal{T}_n$ be the universal set, a set of n distinct attributes, and a set of sub-divided attributes, respectively. Then a pair $(\mathfrak{S}, \mathcal{R})$ is said to be q-ROHSs over χ . When $\mathfrak{S} : \mathcal{R} \rightarrow \mathcal{L}_{q-\text{ROHSs}} \chi$ for ease, we write $\mathcal{R} = \{x^p \times x^q \times \dots \times x^r\}$ such that $0 \leq p, q, r \leq n$ and x^i an element of $\mathcal{R}$. It can be defined and represented by Eq. (4):

$$(\mathfrak{S}, \mathcal{R}) = \left\{ \left\{ \left(x^i, \ell_{\mu(\mathfrak{S})}(x^i), \ell_{\nu(\mathfrak{S})}(x^i) \right) \mid x^i \in \mathcal{R} \right\} \text{ such that } \ell_{\mu} \text{ and } \ell_{\nu} \in \mathcal{E}(\mathcal{R})_{[0,t]} \right\} \quad (4)$$

where, ℓ_{μ} and ℓ_{ν} denotes the degree of linguistic membership and linguistic non-membership of $x^i \in \mathcal{R}$, and holds the restriction that $0 \leq \left(\ell_{\mu(\mathfrak{S})}(x^i) \right)^q + \left(\ell_{\nu(\mathfrak{S})}(x^i) \right)^q \leq 1$, and the degree of hesitancy can be calculated as $\xi_{(\mathfrak{S})}(x^i) = \left| \left(t^q - \left(\mu_{(\mathfrak{S})}(x^i) \right)^q - \left(\nu_{(\mathfrak{S})}(x^i) \right)^q \right)^{\frac{1}{q}} \right|$ for $q \geq 1$.

Example 1: To illustrate the theoretical framework introduced in Definition 5, consider $\chi = \{\text{Crop1}, \text{Crop2}, \text{Crop3}\}$, as a set of crops.

$a_n = \{ \text{Temperature, Humidity, Soil Quality} \}$

$\mathcal{T}_n = \{ \text{Low, Medium, High} \}$

$R = \{x^1 = (\text{Low Temp, Medium Humidity, Good Soil}) \ x^2 = (\text{High Temp, High Humidity, Average Soil}) \}$

Then a pair $(\mathfrak{S}_1, \mathcal{R})$ is said to be q-ROHSs over χ .

Simplification:

Assume $x^1 = \{(\text{Low Temp, Medium Humidity, Good Soil})\}$ with $\ell_{\mu_{(\mathfrak{S}_1)}(x^1)} = 0.6$ and $\ell_{v_{(\mathfrak{S}_1)}(x^1)} = 0.3$ then the constraints $0 \leq (0.6)^2 + (0.3)^2 \leq 1$ for $q = 2$ holds.

The degree of hesitancy for $t = 1$ is:

$$\xi_{(\mathfrak{S}_1)}(x^1) = |1^2 - (0.6)^2 - (0.3)^2|^{\frac{1}{2}} = 0.741$$

$$(\mathfrak{S}_1, \mathcal{R}) = \{ \{ (x^1, 0.6, 0.3, \xi_{(\mathfrak{S}_1)} = 0.741) \mid x^1 \in \mathcal{R} \} \text{ such that } \ell_\mu \text{ and } \ell_\nu \in \mathcal{E}(\mathcal{R})_{[0,1]} \}$$

-Membership (ℓ_μ): Reflects how well the attribute combination fits the ideal scenario.

-Non-Membership (ℓ_ν): Indicates the degree of incompatibility with the ideal.

-Hesitancy (ξ): Measures uncertainty about the classification.

Definition 5. In the above Eq. (4), If ℓ_μ and ℓ_ν of $x^i \in \mathcal{R}$ assigned with fuzzy values,

$$(\mathfrak{S}, \mathcal{R}) = \left\{ \left\{ \left(x^i, \ell_{\mu_{(\mathfrak{S})}(x^i)}, \ell_{\nu_{(\mathfrak{S})}(x^i)} \right) \mid x^i \in \mathcal{R} \right\} \text{ such that } \ell_\mu \text{ and } \ell_\nu \in [0, 1] \right\} \quad (5)$$

In Eq. (5), ℓ_μ and ℓ_ν denotes the degree of fuzzy membership and fuzzy non-membership of $x^i \in \mathcal{R}$ and holds the restriction that $0 \leq \left(\ell_{\mu_{(\mathfrak{S})}(x^i)} \right)^q + \left(\ell_{\nu_{(\mathfrak{S})}(x^i)} \right)^q \leq 1$ and the degree of hesitancy can be calculated as;

$\xi_{(\mathfrak{S})}(x^i) = |t^q - (\mu_{(\mathfrak{S})}(x^i))^q - (v_{(\mathfrak{S})}(x^i))^q|^{\frac{1}{q}}$ for $q \geq 1$. Then it is said to be Lq-ROFHSs.

Example 2: To illustrate the theoretical framework introduced in Definition 6, consider data from Example 1, $x^2 = \{(\text{High Temp, High Humidity, Average Soil})\}$ with $\ell_{\mu_{(\mathfrak{S}_1)}(x^2)} = 0.7$ and $\ell_{v_{(\mathfrak{S}_1)}(x^2)} = 0.4$ then the constraints $0 \leq (0.7)^2 + (0.4)^2 \leq 1$ for $q = 2$ holds.

The degree of hesitancy for $t = 1$ is:

$$\xi_{(\mathfrak{S}_1)}(x^2) = |1^2 - (0.7)^2 - (0.4)^2|^{\frac{1}{2}} = 0.592$$

$$(\mathfrak{S}_1, \mathcal{R}) = \{ \{ (x^2, 0.7, 0.4, \xi_{(\mathfrak{S}_1)} = 0.592) \mid x^2 \in \mathcal{R} \} \text{ such that } \ell_\mu \text{ and } \ell_\nu \in \mathcal{E}(\mathcal{R})_{[0,1]} \}$$

This example demonstrates how fuzzy values are incorporated to express membership and nonmembership degrees, along with hesitancy under Lq-ROFHSs.

3.2 Informational Energies and Correlation Coefficients

Definition 6. Consider $\mathfrak{S}_1 = \left\{ \left(x^i, \ell_{\mu_{(\mathfrak{S}_1)}(x^i)}, \ell_{\nu_{(\mathfrak{S}_1)}(x^i)} \right) \mid x^i \in \mathcal{R} \right\}$ be a Lq-ROFHSs set, where $\mathfrak{S}_1 \in \mathcal{E}(\mathcal{R})_{[0,t]}$.

The informational energy $e(\mathfrak{S}_1)$ of the Lq-ROFHSs $\mathfrak{S}_1$ is defined as follows:

$$e(\mathfrak{S}_1) = \frac{1}{n * t^{2*q}} \sum_{i=1}^n \left(\left(\mu_{(\mathfrak{S}_1)}(x^i) \right)^{2q} + \left(v_{(\mathfrak{S}_1)}(x^i) \right)^{2q} + \left(\xi_{(\mathfrak{S}_1)}(x^i) \right)^{2q} \right) \quad (6)$$

In Eq. (6) $e(\mathfrak{S}_1)$ is non - negative and $e(\mathfrak{S}_1) \leq 1$, and $\xi_{(\mathfrak{S}_1)}(x^i) = |t^q - (\mu_{(\mathfrak{S}_1)}(x^i))^q - (v_{(\mathfrak{S}_1)}(x^i))^q|^{\frac{1}{q}}$ and $q \geq 1$.

Example 3: To illustrate the theoretical framework introduced in Definition 7, and to calculate the information energy for $(\mathfrak{S}_1, \mathcal{R}) = \{x^1, x^2\}$ such that $n = 2$. Consider the rest of the values from example 1.

$$e(\mathfrak{S}_1) = \frac{1}{2 * 1^{2*2}} \sum_{i=1}^2 \left(\left(\mu_{(\mathfrak{S}_1)}(x^i) \right)^{2q} + \left(v_{(\mathfrak{S}_1)}(x^i) \right)^{2q} + \left(\xi_{(\mathfrak{S}_1)}(x^i) \right)^{2q} \right)$$

$$e(\mathfrak{S}_1) = \frac{1}{2} \left((0.6)^4 + (0.3)^4 + (0.741)^4 + (0.7)^4 + (0.3)^4 + (0.592)^4 \right)$$

$$e(\mathfrak{S}_1) = \frac{1}{2}(0.830) = 0.415$$

Similarly, $e(\mathfrak{S}_2) = 0.391$

Definition 7.

Consider $\mathfrak{S}_1 = \left\{ \left(x^i, \ell_{\mu_{(\mathfrak{S}_1)}(x^i)}, \ell_{\nu_{(\mathfrak{S}_1)}(x^i)} \right) \mid x^i \in \mathcal{R} \right\}$ and $\mathfrak{S}_2 = \left\{ \left(x^i, \ell_{\mu_{(\mathfrak{S}_2)}(x^i)}, \ell_{\nu_{(\mathfrak{S}_2)}(x^i)} \right) \mid x^i \in \mathcal{R} \right\}$ be two Lq-ROFHSs, where $\mathfrak{S}_1, \mathfrak{S}_2 \in \mathcal{E}(\mathcal{R})_{[0,t]}$.

The correlation $\mathcal{C}(\mathfrak{S}_1, \mathfrak{S}_2)$ between the two Lq-ROFHSs $\mathfrak{S}_1$ and $\mathfrak{S}_2$ is mathematically defined by Eq. (7) as follows:

$$\mathcal{C}(\mathfrak{J}_1, \mathfrak{J}_2) = \frac{1}{n * t^{2*q}} \sum_{i=1}^n \left[\left((\mu_{(\mathfrak{J}_1)}(x^i))^q (\mu_{(\mathfrak{J}_2)}(x^i))^q + (v_{(\mathfrak{J}_1)}(x^i))^q (v_{(\mathfrak{J}_2)}(x^i))^q + (\xi_{(\mathfrak{J}_1)}(x^i))^q (\xi_{(\mathfrak{J}_2)}(x^i))^q \right) \right] \quad (7)$$

Such that $-1 \leq \mathcal{C}(\mathfrak{J}_1, \mathfrak{J}_2) \leq 1$,

$$\xi_{(\mathfrak{S}_1)}(x^i) = \left| \left(t^q - (\mu_{(\mathfrak{S}_1)}(x^i))^q - (v_{(\mathfrak{S}_1)}(x^i))^q \right) \right|^{\frac{1}{q}},$$

$$\xi_{(\mathfrak{S}_2)}(x^i) = \left| \left(t^q - (\mu_{(\mathfrak{S}_2)}(x^i))^q - (v_{(\mathfrak{S}_2)}(x^i))^q \right) \right|^{\frac{1}{q}} \text{ and } q \geq 1.$$

The proposed correlation satisfies the following properties:

I. $\mathcal{C}(\mathfrak{J}_1, \mathfrak{J}_2) = e(\mathfrak{J}_2)$

II. $\mathcal{C}(\mathfrak{J}_1, \mathfrak{J}_2) = \mathcal{C}(\mathfrak{J}_2, \mathfrak{J}_1)$

Example 4: To illustrate the theoretical framework introduced in Definition 8, consider $\mathfrak{S}_1 = \{(x^1, x^2)\}$ and $\mathfrak{S}_2 = (x^1, x^2)$ be two Lq-ROFHSs, where $\mathfrak{S}_1$, and $\mathfrak{S}_2 \in \mathcal{E}(\mathcal{R})_{[0,1]}$. Using $n = 2, t = 1$, and $q = 2$ in Eq. (6), we get the results. Consider the values from example 1 and example 2 for mapping $\mathfrak{J}_1$

$$(\mathfrak{S}_1, \mathcal{R}) = \left\{ \left\{ (x^1, 0.6, 0.3, \xi_{(\mathfrak{S}_1)} = 0.741) \mid x^1 \in \mathcal{R} \right\} \text{ such that } \ell_\mu \text{ and } \ell_\nu \in \mathcal{E}(\mathcal{R})_{[0,1]} \right\}$$

$$(\mathfrak{S}_1, \mathcal{R}) = \left\{ \left\{ (x^2, 0.7, 0.4, \xi_{(\mathfrak{S}_1)} = 0.592) \mid x^2 \in \mathcal{R} \right\} \text{ such that } \ell_\mu \text{ and } \ell_\nu \in \mathcal{E}(\mathcal{R})_{[0,1]} \right\}$$

For mapping $\mathfrak{J}_2$ consider

$$(\mathfrak{S}_2, \mathcal{R}) = \left\{ \left\{ (x^1, 0.5, 0.4, \xi_{(\mathfrak{S}_2)} = 0.768) \mid x^1 \in \mathcal{R} \right\} \text{ such that } \ell_\mu \text{ and } \ell_\nu \in \mathcal{E}(\mathcal{R})_{[0,1]} \right\}$$

$$(\mathfrak{J}_2, \mathcal{R}) = \left\{ \left\{ (x^2, 0.6, 0.5, \xi_{(\mathfrak{S}_2)} = 0.624) \mid x^2 \in \mathcal{R} \right\} \text{ such that } \ell_\mu \text{ and } \ell_\nu \in \mathcal{E}(\mathcal{R})_{[0,1]} \right\}$$

To find the $\mathcal{C}(\mathfrak{S}_1, \mathfrak{S}_2)$ substitute values in Eq. (7).

$$\mathcal{C}(\mathfrak{J}_1, \mathfrak{J}_2) = \frac{1}{2}(0.429 + 0.353) = 0.391$$

The results clearly satisfy properties (I) and (II).

I. $\mathcal{C}(\mathfrak{J}_1, \mathfrak{J}_2) = e(\mathfrak{J}_2)$ i.e. $\mathcal{C}(\mathfrak{J}_1, \mathfrak{J}_2) = 0.391 = e(\mathfrak{J}_2)$

II. $\mathcal{C}(\mathfrak{J}_1, \mathfrak{J}_2) = \mathcal{C}(\mathfrak{J}_2, \mathfrak{J}_1)$ hold. The correlation formula is symmetric because the terms in Eq. (7) are multiplicative, and multiplication is commutative.

Definition 8.

Consider $\mathfrak{S}_1 = \left\{ \left(x^i, \ell_{\mu_{(\mathfrak{S}_1)}(x^i)}, \ell_{\nu_{(\mathfrak{S}_1)}(x^i)} \right) \mid x^i \in \mathcal{R} \right\}$ and $\mathfrak{S}_2 = \left\{ \left(x^i, \ell_{\mu_{(\mathfrak{S}_2)}(x^i)}, \ell_{\nu_{(\mathfrak{S}_2)}(x^i)} \right) \mid x^i \in \mathcal{R} \right\}$ be two Lq-ROFHSs, where $\mathfrak{S}_1, \mathfrak{S}_2 \in \mathcal{E}(\mathcal{R})_{[0,t]}$.

The correlation $\mathcal{C}(\mathfrak{S}_1, \mathfrak{S}_2)$ between the two Linguistic q-Rung Orthopair Fuzzy Hyperspft Numbers (Lq-ROFHSs) $\mathfrak{J}_1$, and $\mathfrak{J}_2$ is mathematically defined by Eq. (8) as follows:

$$\mathfrak{C}(\mathfrak{S}_1, \mathfrak{S}_2) = \frac{\mathcal{C}(\mathfrak{S}_1, \mathfrak{S}_2)}{\max \langle e(\mathfrak{S}_1), e(\mathfrak{S}_2) \rangle} \quad (8)$$

Such that $-1 \leq \mathcal{C}(\mathfrak{S}_1, \mathfrak{S}_2) \leq 1$.

Example 5: To illustrate the theoretical framework introduced in Definition 9 and determine $\mathfrak{C}(\mathfrak{S}_1, \mathfrak{S}_2)$, Eq. (8) is applied using the values obtained from all the preceding examples.

$$\mathfrak{C}(\mathfrak{S}_1, \mathfrak{S}_2) = \frac{\mathcal{C}(\mathfrak{S}_1, \mathfrak{S}_2)}{\max \langle e(\mathfrak{S}_1), e(\mathfrak{S}_2) \rangle}$$

$$\mathfrak{C}(\mathfrak{S}_1, \mathfrak{S}_2) = \frac{0.391}{\max \langle 0.415, 0.391 \rangle}$$

$$\mathfrak{C}(\mathfrak{S}_1, \mathfrak{S}_2) = \frac{0.391}{0.806} = 0.4851$$

Theorem 1.

Consider $\mathfrak{S}_1 = \left\{ \left(x^i, \ell_{\mu_{(\mathfrak{S}_1)}(x^i)}, \ell_{\nu_{(\mathfrak{S}_1)}(x^i)} \right) \mid x^i \in \mathcal{R} \right\}$ and $\mathfrak{S}_2 = \left\{ \left(x^i, \ell_{\mu_{(\mathfrak{S}_2)}(x^i)}, \ell_{\nu_{(\mathfrak{S}_2)}(x^i)} \right) \mid x^i \in \mathcal{R} \right\}$ be two Lq-ROFHSs, where $\mathfrak{S}_1, \mathfrak{S}_2 \in \mathcal{E}(\mathcal{R})_{[0,t]}$. The proposed a Correlation Coefficient $\mathfrak{C}(\mathfrak{S}_1, \mathfrak{S}_2)$ between the Lq-ROFHSs $\mathfrak{S}_1$, and $\mathfrak{S}_2$ defined in Eq. (8) satisfies the following conditions:

- I. $\mathfrak{C}(\mathfrak{S}_1, \mathfrak{S}_2) = \mathfrak{C}(\mathfrak{S}_2, \mathfrak{S}_1)$
- II. The $\mathfrak{C}(\mathfrak{S}_2, \mathfrak{S}_1)$ will be non - negative and $\mathfrak{C}(\mathfrak{S}_2, \mathfrak{S}_1) \leq 1$
- III. $\mathfrak{S}_1 = \mathfrak{S}_2 \Rightarrow \mathfrak{C}(\mathfrak{S}_1, \mathfrak{S}_1) = 1$

Proof. Let $\mathfrak{S}_1 = \left\{ \left(x^i, \ell_{\mu_{(\mathfrak{S}_1)}(x^i)}, \ell_{\nu_{(\mathfrak{S}_1)}(x^i)} \right) \mid x^i \in \mathcal{R} \right\}$ and $\mathfrak{S}_2 = \left\{ \left(x^i, \ell_{\mu_{(\mathfrak{S}_2)}(x^i)}, \ell_{\nu_{(\mathfrak{S}_2)}(x^i)} \right) \mid x^i \in \mathcal{R} \right\}$ be two Lq-ROFHSs, where $\mathfrak{S}_1, \mathfrak{S}_2 \in \mathcal{E}(\mathcal{R})_{[0,t]}$.

I. Consider definition 8,

$$\mathfrak{C}(\mathfrak{S}_1, \mathfrak{S}_2) = \frac{\mathcal{C}(\mathfrak{S}_1, \mathfrak{S}_2)}{\max \langle e(\mathfrak{S}_1), e(\mathfrak{S}_2) \rangle}$$

$$\begin{aligned} \mathfrak{C}(\mathfrak{S}_1, \mathfrak{S}_2) &= \frac{\sum_{i=1}^n \left[\left((\mu_{(\mathfrak{S}_1)}(x^i))^q (\mu_{(\mathfrak{S}_2)}(x^i))^q + (\nu_{(\mathfrak{S}_1)}(x^i))^q (\nu_{(\mathfrak{S}_2)}(x^i))^q + (\xi_{(\mathfrak{S}_1)}(x^i))^q (\xi_{(\mathfrak{S}_2)}(x^i))^q \right) \right]}{\max \left\langle \sum_{i=1}^n \left((\mu_{(\mathfrak{S}_1)}(x^i))^{2q} + (\nu_{(\mathfrak{S}_1)}(x^i))^{2q} + (\xi_{(\mathfrak{S}_1)}(x^i))^{2q} \right), \sum_{i=1}^n \left((\mu_{(\mathfrak{S}_2)}(x^i))^{2q} + (\nu_{(\mathfrak{S}_2)}(x^i))^{2q} + (\xi_{(\mathfrak{S}_2)}(x^i))^{2q} \right) \right\rangle} \\ &= \frac{\sum_{i=1}^n \left[\left((\mu_{(\mathfrak{S}_2)}(x^i))^q (\mu_{(\mathfrak{S}_1)}(x^i))^q + (\nu_{(\mathfrak{S}_2)}(x^i))^q (\nu_{(\mathfrak{S}_1)}(x^i))^q + (\xi_{(\mathfrak{S}_2)}(x^i))^q (\xi_{(\mathfrak{S}_1)}(x^i))^q \right) \right]}{\max \left\langle \sum_{i=1}^n \left((\mu_{(\mathfrak{S}_2)}(x^i))^{2q} + (\nu_{(\mathfrak{S}_2)}(x^i))^{2q} + (\xi_{(\mathfrak{S}_2)}(x^i))^{2q} \right), \sum_{i=1}^n \left((\mu_{(\mathfrak{S}_1)}(x^i))^{2q} + (\nu_{(\mathfrak{S}_1)}(x^i))^{2q} + (\xi_{(\mathfrak{S}_1)}(x^i))^{2q} \right) \right\rangle} = \mathfrak{C}(\mathfrak{S}_2, \mathfrak{S}_1) \end{aligned}$$

II. The non-negativity is straightforward. It therefore remains only to prove that $\mathfrak{C}(\mathfrak{S}_1, \mathfrak{S}_2) \leq 1$

$$\mathfrak{C}(\mathfrak{S}_2, \mathfrak{S}_1) = \sum_{i=1}^n \left[\left((\mu_{(\mathfrak{S}_1)}(x^i))^q (\mu_{(\mathfrak{S}_2)}(x^i))^q + (\nu_{(\mathfrak{S}_1)}(x^i))^q (\nu_{(\mathfrak{S}_2)}(x^i))^q + (\xi_{(\mathfrak{S}_1)}(x^i))^q (\xi_{(\mathfrak{S}_2)}(x^i))^q \right) \right]$$

$$\begin{aligned} \mathfrak{C}(\mathfrak{S}_2, \mathfrak{S}_1) &= \left[\left\{ (\mu_{(\mathfrak{S}_1)}(x^1))^q (\mu_{(\mathfrak{S}_2)}(x^1))^q + (\nu_{(\mathfrak{S}_1)}(x^1))^q (\nu_{(\mathfrak{S}_2)}(x^1))^q + (\xi_{(\mathfrak{S}_1)}(x^1))^q (\xi_{(\mathfrak{S}_2)}(x^1))^q \right\} \right. \\ &\quad + \left\{ (\mu_{(\mathfrak{S}_1)}(x^2))^q (\mu_{(\mathfrak{S}_2)}(x^2))^q + (\nu_{(\mathfrak{S}_1)}(x^2))^q (\nu_{(\mathfrak{S}_2)}(x^2))^q + (\xi_{(\mathfrak{S}_1)}(x^2))^q (\xi_{(\mathfrak{S}_2)}(x^2))^q \right\} \\ &\quad + \left\{ (\mu_{(\mathfrak{S}_1)}(x^3))^q (\mu_{(\mathfrak{S}_2)}(x^3))^q + (\nu_{(\mathfrak{S}_1)}(x^3))^q (\nu_{(\mathfrak{S}_2)}(x^3))^q + (\xi_{(\mathfrak{S}_1)}(x^3))^q (\xi_{(\mathfrak{S}_2)}(x^3))^q \right\} + \\ &\quad \left. \dots + \left\{ (\mu_{(\mathfrak{S}_1)}(x^n))^q (\mu_{(\mathfrak{S}_2)}(x^n))^q + (\nu_{(\mathfrak{S}_1)}(x^n))^q (\nu_{(\mathfrak{S}_2)}(x^n))^q + (\xi_{(\mathfrak{S}_1)}(x^n))^q (\xi_{(\mathfrak{S}_2)}(x^n))^q \right\} \right] \end{aligned}$$

Using Cauchy-Schwarz inequality:

$$\begin{aligned} (x^1 y^1 + x^2 y^2 + x^3 y^3 + \dots + x^n y^n)^2 &\leq \left[(x^1)^2 + (x^2)^2 + \dots + (x^n)^2 \right] \times \left[(y^1)^2 + (y^2)^2 + \dots + (y^n)^2 \right] \\ \mathfrak{C}(\mathfrak{S}_2, \mathfrak{S}_1)^2 &= \left[\sum_{i=1}^n \left((\mu_{(\mathfrak{S}_1)}(x^i))^q (\mu_{(\mathfrak{S}_2)}(x^i))^q + (\nu_{(\mathfrak{S}_1)}(x^i))^q (\nu_{(\mathfrak{S}_2)}(x^i))^q + (\xi_{(\mathfrak{S}_1)}(x^i))^q (\xi_{(\mathfrak{S}_2)}(x^i))^q \right) \right]^2 \\ &\leq \left[\sum_{i=1}^n \left((\mu_{(\mathfrak{S}_1)}(x^i))^q + (\nu_{(\mathfrak{S}_1)}(x^i))^q + (\xi_{(\mathfrak{S}_1)}(x^i))^q \right) \right]^2 \times \left[\sum_{i=1}^n \left((\mu_{(\mathfrak{S}_2)}(x^i))^q + (\nu_{(\mathfrak{S}_2)}(x^i))^q + (\xi_{(\mathfrak{S}_2)}(x^i))^q \right) \right]^2 \\ &\leq \left[\sum_{i=1}^n \left((\mu_{(\mathfrak{S}_1)}(x^i))^{2q} + (\nu_{(\mathfrak{S}_1)}(x^i))^{2q} + (\xi_{(\mathfrak{S}_1)}(x^i))^{2q} \right) \right] \times \left[\sum_{i=1}^n \left((\mu_{(\mathfrak{S}_2)}(x^i))^{2q} + (\nu_{(\mathfrak{S}_2)}(x^i))^{2q} + (\xi_{(\mathfrak{S}_2)}(x^i))^{2q} \right) \right] \\ &= e(\mathfrak{S}_1) \times e(\mathfrak{S}_2) \\ &\because e(\mathfrak{S}_1) \text{ and } e(\mathfrak{S}_2) \leq 1 \\ \therefore \mathfrak{C}(\mathfrak{S}_2, \mathfrak{S}_1) &\leq \max \langle e(\mathfrak{S}_1), e(\mathfrak{S}_2) \rangle \Rightarrow \mathfrak{C}(\mathfrak{S}_2, \mathfrak{S}_1) \leq 1. \end{aligned}$$

III. To prove $\mathfrak{S}_1 \times \mathfrak{S}_2 \Rightarrow \mathfrak{C}(\mathfrak{S}_1, \mathfrak{S}_2) = 1$ Let $\mathfrak{S}_1 = \mathfrak{S}_2$ be two equals Lq-ROFHSs, then,

$$\ell_{\mu_{(\mathfrak{S}_1)}(x^i)} = \ell_{\mu_{(\mathfrak{S}_2)}(x^i)}, \ell_{\nu_{(\mathfrak{S}_1)}(x^i)} = \ell_{\nu_{(\mathfrak{S}_2)}(x^i)} \text{ and } \ell_{\xi_{(\mathfrak{S}_1)}(x^i)} = \ell_{\xi_{(\mathfrak{S}_2)}(x^i)} \text{ where } \mathfrak{S}_1, \mathfrak{S}_2 \in \mathcal{E}(\mathcal{R})_{[0,t]}$$

Then using Eq. (8)

$$\mathfrak{C}(\mathfrak{S}_1, \mathfrak{S}_2) = \frac{\mathcal{C}(\mathfrak{S}_1, \mathfrak{S}_2)}{\max \langle e(\mathfrak{S}_1), e(\mathfrak{S}_2) \rangle}$$

put $\mathfrak{S}_2 = \mathfrak{S}_1$,

$$\mathfrak{C}(\mathfrak{S}_1, \mathfrak{S}_2) = \frac{\sum_{i=1}^n [(\mu_{(\mathfrak{S}_1)}(x^i))^q (\mu_{(\mathfrak{S}_1)}(x^i))^q + (v_{(\mathfrak{S}_1)}(x^i))^q (v_{(\mathfrak{S}_1)}(x^i))^q + (\xi_{(\mathfrak{S}_1)}(x^i))^q (\xi_{(\mathfrak{S}_1)}(x^i))^q]}{\max \left\langle \sum_{i=1}^n \left((\mu_{(\mathfrak{S}_1)}(x^i))^{2q} + (v_{(\mathfrak{S}_1)}(x^i))^{2q} + (\xi_{(\mathfrak{S}_1)}(x^i))^{2q} \right), \sum_{i=1}^n \left((\mu_{(\mathfrak{S}_1)}(x^i))^{2q} + (v_{(\mathfrak{S}_1)}(x^i))^{2q} + (\xi_{(\mathfrak{S}_1)}(x^i))^{2q} \right) \right\rangle}$$

$$\mathfrak{C}(\mathfrak{S}_1, \mathfrak{S}_2) = 1$$

Definition 9. Consider $\mathfrak{S}_1 = \left\{ \left(x^i, \ell_{\mu_{(\mathfrak{S}_1)}(x^i)}, \ell_{\nu_{(\mathfrak{S}_1)}(x^i)} \right) \mid x^i \in \mathcal{R} \right\}$ be a Lq-ROFHSs set, where $\mathfrak{S}_1 \in \mathcal{E}(\mathcal{R})_{[0,t]}$.

A weighted informational energy $e_\omega(\mathfrak{S}_1)$ is proposed for the Lq-ROFHSs $\mathfrak{S}_1$ and is mathematically defined by Eq. (9) as follows:

$$e_\omega(\mathfrak{S}_1) = \frac{1}{t^{2*q}} \sum_{i=1}^n \left(\omega_i (\mu_{(\mathfrak{S}_1)}(x^i))^{2q} + (v_{(\mathfrak{S}_1)}(x^i))^{2q} + (\xi_{(\mathfrak{S}_1)}(x^i))^{2q} \right) \quad (9)$$

where, ω_i is the weight of the element x^i . Such that $e_\omega(\mathfrak{S}_1)$ is $-1 \leq e_\omega(\mathfrak{S}_1) \leq 1$, and $x^i, \omega_i \geq 1$ with $\sum_{i=1}^n \omega_i = 1$.

$$\xi_{(\mathfrak{S}_1)}(x^i) = \left| \left(t^q - (\mu_{(\mathfrak{S}_1)}(x^i))^q - (v_{(\mathfrak{S}_1)}(x^i))^q \right)^{\frac{1}{q}} \right| \text{ and } q \geq 1.$$

Definition 10.

Consider $\mathfrak{S}_1 = \left\{ \left(x^i, \ell_{\mu_{(\mathfrak{S}_1)}(x^i)}, \ell_{\nu_{(\mathfrak{S}_1)}(x^i)} \right) \mid x^i \in \mathcal{R} \right\}$ and $\mathfrak{S}_2 = \left\{ \left(x^i, \ell_{\mu_{(\mathfrak{S}_2)}(x^i)}, \ell_{\nu_{(\mathfrak{S}_2)}(x^i)} \right) \mid x^i \in \mathcal{R} \right\}$ be two Lq-ROFHSs, where $\mathfrak{S}_1, \mathfrak{S}_2 \in \mathcal{E}(\mathcal{R})_{[0,t]}$. The weighted correlation $\mathcal{C}_\omega(\mathfrak{S}_1, \mathfrak{S}_2)$ between the two Lq-ROFHSs $\mathfrak{S}_1$ and $\mathfrak{S}_2$ is defined mathematically by Eq. (10) as follows:

$$\mathcal{C}_\omega(\mathfrak{S}_1, \mathfrak{S}_2) = \frac{1}{t^{2*q}} \sum_{i=1}^n \left[\omega_i \left((\mu_{(\mathfrak{S}_1)}(x^i))^q (\mu_{(\mathfrak{S}_2)}(x^i))^q + (\nu_{(\mathfrak{S}_1)}(x^i))^q (\nu_{(\mathfrak{S}_2)}(x^i))^q + (\xi_{(\mathfrak{S}_1)}(x^i))^q (\xi_{(\mathfrak{S}_2)}(x^i))^q \right) \right] \quad (10)$$

Such that $\mathcal{C}_\omega(\mathfrak{S}_1, \mathfrak{S}_2)$ is $-1 \leq \mathcal{C}_\omega(\mathfrak{S}_1, \mathfrak{S}_2) \leq 1$, where ω_i is the weight of the element x^i , and $x^i, \omega_i \geq 1$ with $\sum_{i=1}^n \omega_i = 1$,

$$\xi_{(\mathfrak{S}_1)}(x^i) = \left| \left(t^q - (\mu_{(\mathfrak{S}_1)}(x^i))^q - (v_{(\mathfrak{S}_1)}(x^i))^q \right)^{\frac{1}{q}} \right|,$$

$$\xi_{(\mathfrak{S}_2)}(x^i) = \left| \left(t^q - (\mu_{(\mathfrak{S}_2)}(x^i))^q - (v_{(\mathfrak{S}_2)}(x^i))^q \right)^{\frac{1}{q}} \right|,$$

and $q \geq 1$.

It will satisfy the following properties:

$$\text{I. } \mathcal{C}_\omega(\mathfrak{S}_1, \mathfrak{S}_2) = e_\omega(\mathfrak{S}_1)$$

$$\text{II. } \mathcal{C}_\omega(\mathfrak{S}_1, \mathfrak{S}_2) = \mathcal{C}_\omega(\mathfrak{S}_2, \mathfrak{S}_1)$$

Definition 11.

Consider $\mathfrak{S}_1 = \left\{ \left(x^i, \ell_{\mu_{(\mathfrak{S}_1)}(x^i)}, \ell_{\nu_{(\mathfrak{S}_1)}(x^i)} \right) \mid x^i \in \mathcal{R} \right\}$ and $\mathfrak{S}_2 = \left\{ \left(x^i, \ell_{\mu_{(\mathfrak{S}_2)}(x^i)}, \ell_{\nu_{(\mathfrak{S}_2)}(x^i)} \right) \mid x^i \in \mathcal{R} \right\}$ be two Lq-ROFHSs, where $\mathfrak{S}_1, \mathfrak{S}_2 \in \mathcal{E}(\mathcal{R})_{[0,t]}$. A weighted correlation coefficient $\mathfrak{C}_\omega(\mathfrak{S}_1, \mathfrak{S}_2)$ is proposed for the Lq-ROFHSs $\mathfrak{S}_1$, and $\mathfrak{S}_2$ and is defined by Eq. (11) as follows:

$$\mathfrak{C}_\omega(\mathfrak{S}_1, \mathfrak{S}_2) = \frac{\mathcal{C}_\omega(\mathfrak{S}_1, \mathfrak{S}_2)}{\max \langle e_\omega(\mathfrak{S}_1), e_\omega(\mathfrak{S}_2) \rangle} \quad (11)$$

Such that $\mathfrak{C}_\omega(\mathfrak{S}_1, \mathfrak{S}_2)$ is $-1 \leq \mathfrak{C}_\omega(\mathfrak{S}_1, \mathfrak{S}_2) \leq 1$. Where ω_i is the weight of the element x^i , and $x^i, \omega_i \geq 1$ with $\sum_{i=1}^n \omega_i = 1$.

Example 6: To illustrate the theoretical framework introduced in Definitions 10–12, consider $\tilde{\mathfrak{S}}_1 = \{ (x^1, \ell_1, \ell_2), (x^2, \ell_3, \ell_5), (x^3, \ell_3, \ell_3), (x^4, \ell_6, \ell_2) \}$ and $\mathfrak{S}_2 = (x^1, \ell_2, \ell_3), (x^2, \ell_0, \ell_1), (x^3, \ell_1, \ell_2), (x^4, \ell_2, \ell_2)$ be two Lq-ROFHSs, where $\mathfrak{S}_1$, and $\mathfrak{S}_2 \in \mathcal{E}(\mathcal{R})_{[0,6]}$. Using $t = 5, m = 4, q = 2$ and $\omega_1 = 0.19, \omega_2 = 0.31, \omega_3 = 0.27, \omega_4 = 0.23$, the weighted informational energy is obtained from Eq. (9) as follows:

The following quantities are calculated:

a. Weighted informational energy $e_\omega(\mathfrak{S}_1)$

b. Weighted Correlation $\mathcal{C}_\omega(\mathfrak{S}_1, \mathfrak{S}_2)$

c. Weighted Correlation Coefficient $\mathfrak{C}_\omega(\mathfrak{S}_1, \mathfrak{S}_2)$

(a) Weighted Informational Energy: Using Eq. (9), the weighted informational energy is obtained as follows

$$\begin{aligned}
e_{\omega}(\mathfrak{S}_1) &= \frac{1}{t^{2*q}} \sum_{i=1}^n \left(\omega_i (\mu_{(\mathfrak{S}_1)}(x^i))^{2q} + (v_{(\mathfrak{S}_1)}(x^i))^{2q} + (\xi_{(\mathfrak{S}_1)}(x^i))^{2q} \right) \\
&= \frac{1}{5^{2*2}} \sum_{i=1}^4 \left(\omega_i (\mu_{(\mathfrak{S}_1)}(x^i))^{2*2} + (v_{(\mathfrak{S}_1)}(x^i))^{2*2} + (\xi_{(\mathfrak{S}_1)}(x^i))^{2*2} \right) \\
&= \frac{1}{625} \left(0.19 \left(1^4 + 2^4 + |2^2 - 1^2 - 2^2|^{\frac{1}{2}} \right) + 0.31 \left(3^4 + 5^4 + |2^2 - 3^2 - 5^2|^{\frac{1}{2}} \right) + 0.27 \left(3^4 + 3^4 + |2^2 - 3^2 - 3^2|^{\frac{1}{2}} \right) + 0.23 \left(6^4 + 2^4 + |2^2 - 6^2 - 2^2|^{\frac{1}{2}} \right) \right) \\
&= \frac{1}{625} (0.19(18) + 0.31(711.4771) + 0.27(164) + 0.23(1318)) \\
&= \frac{3.42 + 220.5579 + 44.28 + 303.14}{625} \\
e_{\omega}(\mathfrak{S}_1) &= 0.9142
\end{aligned}$$

(b) Weighted Correlation: Using Eq. (10), the weighted correlation is obtained as follows:

$$\begin{aligned}
e_{\omega}(\mathfrak{S}_2) &= \frac{1}{t^{2*q}} \sum_{i=1}^n \left(\omega_i (\mu_{(\mathfrak{S}_2)}(x^i))^{2q} + (v_{(\mathfrak{S}_2)}(x^i))^{2q} + (\xi_{(\mathfrak{S}_2)}(x^i))^{2q} \right) \\
&= \frac{1}{5^{2*2}} \sum_{i=1}^4 \left(\omega_i (\mu_{(\mathfrak{S}_2)}(x^i))^{2*2} + (v_{(\mathfrak{S}_2)}(x^i))^{2*2} + (\xi_{(\mathfrak{S}_2)}(x^i))^{2*2} \right) \\
&= \frac{1}{625} \left(0.19 \left(2^4 + 3^4 + |2^2 - 2^2 - 3^2|^{\frac{1}{2}} \right) + 0.31 \left(0^4 + 1^4 + |2^2 - 0^2 - 1^2|^{\frac{1}{2}} \right) + 0.27 \left(1^4 + 2^4 + |2^2 - 1^2 - 2^2|^{\frac{1}{2}} \right) + 0.23 \left(2^4 + 2^4 + |2^2 - 2^2 - 2^2|^{\frac{1}{2}} \right) \right) \\
&= \frac{1}{625} (0.19(100) + 0.31(2) + 0.27(18) + 0.23(34)) \\
&= \frac{19 + 0.62 + 4.86 + 7.82}{625} \\
e_{\omega}(\mathfrak{S}_2) &= 0.05168
\end{aligned}$$

(c) Weighted Correlation Coefficient: Using Eq. (11), the weighted correlation coefficient is obtained as follows:

$$\mathfrak{C}_{\omega}(\mathfrak{S}_1, \mathfrak{S}_2) = \frac{\mathcal{C}_{\omega}(\mathfrak{S}_1, \mathfrak{S}_2)}{\max \langle e_{\omega}(\mathfrak{S}_1), e_{\omega}(\mathfrak{S}_2) \rangle} = \frac{0.5168}{\max \langle 0.9142, 0.05168 \rangle} = \frac{0.0516}{0.9142} = 0.0564$$

Theorem 2: Consider two Lq-ROFHSs as defined above. The weighted correlation coefficient defined in Eq. (11) satisfies the following properties:

- I. $\mathfrak{C}_{\omega}(\mathfrak{S}_1, \mathfrak{S}_2) = \mathfrak{C}_{\omega}(\mathfrak{S}_2, \mathfrak{S}_1)$
- II. The $\mathfrak{C}_{\omega}(\mathfrak{S}_1, \mathfrak{S}_2)$ will be $-1 \leq \mathfrak{C}_{\omega}(\mathfrak{S}_1, \mathfrak{S}_2) \leq 1$
- III. $\mathfrak{S}_1 = \mathfrak{S}_2 \Rightarrow \mathfrak{C}_{\omega}(\mathfrak{S}_1, \mathfrak{S}_2) = 1$

Proof. The proof is the same as in Theorem 1.

Theorem 3. Let $Q_{e_{ij}} = (\alpha_{ij}, \beta_{ij})$ be the collection of q -ROHSs over χ then

$$\begin{aligned}
&q\text{-ROHSs}(Q_{e_{11}}, Q_{e_{12}}, \dots, Q_{e_{nm}}) \\
&= \left\langle \frac{\sqrt[q]{2 \prod_{j=1}^m \left(\prod_{i=1}^n (\alpha_{ij}^q)^{\Xi_i} \right)^{\eta_j}}}{\sqrt[q]{\prod_{j=1}^m \left(\prod_{i=1}^n (2 - \alpha_{ij}^q)^{\Xi_i} \right)^{\eta_j} + \prod_{j=1}^m \left(\prod_{i=1}^n (\alpha_{ij}^q)^{\Xi_i} \right)^{\eta_j}}}, \frac{\sqrt[q]{\prod_{j=1}^m \left(\prod_{i=1}^n (1 + \beta_{ij}^q)^{\Xi_i} \right)^{\eta_j} - \prod_{j=1}^m \left(\prod_{i=1}^n (1 - \beta_{ij}^q)^{\Xi_i} \right)^{\eta_j}}}{\sqrt[q]{\prod_{j=1}^m \left(\prod_{i=1}^n (1 + \beta_{ij}^q)^{\Xi_i} \right)^{\eta_j} + \prod_{j=1}^m \left(\prod_{i=1}^n (1 - \beta_{ij}^q)^{\Xi_i} \right)^{\eta_j}}} \right\rangle
\end{aligned}$$

where, $\Xi_i > 0, \eta_j > 0$ denotes the weights of experts and attributes, and $\sum_{i=1}^n \Xi_i = \sum_{j=1}^m \eta_j = 1$.

Proof. This theorem is proved by mathematical induction. Let $n = 1$, and $\Xi_i = 1$.

$$\begin{aligned}
q\text{-ROHSs}(Q_{e_{11}}, Q_{e_{12}}, \dots, Q_{e_{nm}}) &= \otimes_{j=1}^m (Q_{e_{1j}})^{\eta_j} = \left\langle \frac{\sqrt[q]{2 \prod_{j=1}^m (\alpha_{1j}^q)^{\eta_j}}}{\sqrt[q]{\prod_{j=1}^m (2 - \alpha_{1j}^q)^{\eta_j} + \prod_{j=1}^m (\alpha_{1j}^q)^{\eta_j}}}, \frac{\sqrt[q]{\prod_{j=1}^m (1 + \beta_{1j}^q)^{\eta_j} - \prod_{j=1}^m (1 - \beta_{1j}^q)^{\eta_j}}}{\sqrt[q]{\prod_{j=1}^m (1 + \beta_{1j}^q)^{\eta_j} + \prod_{j=1}^m (1 - \beta_{1j}^q)^{\eta_j}}} \right\rangle \\
&= \left\langle \frac{\sqrt[q]{2 \prod_{j=1}^m \left(\prod_{i=1}^1 (\alpha_{ij}^q)^{\Xi_i} \right)^{\eta_j}}}{\sqrt[q]{\prod_{j=1}^m \left(\prod_{i=1}^1 (2 - \alpha_{ij}^q)^{\Xi_i} \right)^{\eta_j} + \prod_{j=1}^m \left(\prod_{i=1}^1 (\alpha_{ij}^q)^{\Xi_i} \right)^{\eta_j}}}, \frac{\sqrt[q]{\prod_{j=1}^m \left(\prod_{i=1}^1 (1 + \beta_{ij}^q)^{\Xi_i} \right)^{\eta_j} - \prod_{j=1}^m \left(\prod_{i=1}^1 (1 - \beta_{ij}^q)^{\Xi_i} \right)^{\eta_j}}}{\sqrt[q]{\prod_{j=1}^m \left(\prod_{i=1}^1 (1 + \beta_{ij}^q)^{\Xi_i} \right)^{\eta_j} + \prod_{j=1}^m \left(\prod_{i=1}^1 (1 - \beta_{ij}^q)^{\Xi_i} \right)^{\eta_j}}} \right\rangle
\end{aligned}$$

For $m = 1$, we get $\eta_j = 1$.

$$\begin{aligned}
q\text{-ROHSs}(Q_{e_{11}}, Q_{e_{12}}, \dots, Q_{e_{nm}}) &= \otimes_{i=1}^n (Q_{e_{i1}})^{\Xi_i} = \left\langle \frac{\sqrt[q]{2 \prod_{i=1}^n (\alpha_{ij}^q)^{\Xi_i}}}{\sqrt[q]{\prod_{i=1}^n (2 - \alpha_{ij}^q)^{\Xi_i} + \prod_{i=1}^n (\alpha_{ij}^q)^{\Xi_i}}}, \frac{\sqrt[q]{\prod_{i=1}^n (1 + \beta_{ij}^q)^{\Xi_i} - \prod_{i=1}^n (1 - \beta_{ij}^q)^{\Xi_i}}}{\sqrt[q]{\prod_{i=1}^n (1 + \beta_{ij}^q)^{\Xi_i} + \prod_{i=1}^n (1 - \beta_{ij}^q)^{\Xi_i}}} \right\rangle \\
&= \left\langle \frac{\sqrt[q]{2 \prod_{j=1}^1 \left(\prod_{i=1}^n (\alpha_{ij}^q)^{\Xi_i} \right)^{\eta_j}}}{\sqrt[q]{\prod_{j=1}^1 \left(\prod_{i=1}^n (2 - \alpha_{ij}^q)^{\Xi_i} \right)^{\eta_j} + \prod_{j=1}^1 \left(\prod_{i=1}^n (\alpha_{ij}^q)^{\Xi_i} \right)^{\eta_j}}}, \frac{\sqrt[q]{\prod_{j=1}^1 \left(\prod_{i=1}^n (1 + \beta_{ij}^q)^{\Xi_i} \right)^{\eta_j} - \prod_{j=1}^1 \left(\prod_{i=1}^n (1 - \beta_{ij}^q)^{\Xi_i} \right)^{\eta_j}}}{\sqrt[q]{\prod_{j=1}^1 \left(\prod_{i=1}^n (1 + \beta_{ij}^q)^{\Xi_i} \right)^{\eta_j} + \prod_{j=1}^1 \left(\prod_{i=1}^n (1 - \beta_{ij}^q)^{\Xi_i} \right)^{\eta_j}}} \right\rangle
\end{aligned}$$

So, the statement holds for $n = 1$ and $m = 1$.

Suppose for $n = \omega_2, m = \omega_1 + 1$ and for $n = \omega_2 + 1, m = \omega_1$.

$$q - \text{ROHSS}(Q_{e_{11}}, Q_{e_{12}}, \dots, Q_{e_{nm}}) = \otimes_{j=1}^{\omega_1+1} \left(\otimes_{i=1}^{\omega_2} (Q_{e_{ij}})^{\Xi_i} \right)^{\eta_j}$$

$$\left\langle \sqrt[q]{\frac{2 \prod_{j=1}^{\omega_1+1} \left(\prod_{i=1}^{\omega_2} (a_{ij}^q)^{\Xi_i} \right)^{\eta_j}}{\prod_{j=1}^{\omega_1+1} \left(\prod_{i=1}^{\omega_2} (2 - a_{ij}^q)^{\Xi_i} \right)^{\eta_j} + \prod_{j=1}^{\omega_1+1} \left(\prod_{i=1}^{\omega_2} (a_{ij}^q)^{\Xi_i} \right)^{\eta_j}}}, \sqrt[q]{\frac{\prod_{j=1}^{\omega_1+1} \left(\prod_{i=1}^{\omega_2} (1 + \beta_{ij}^q)^{\Xi_i} \right)^{\eta_j} - \prod_{j=1}^{\omega_1+1} \left(\prod_{i=1}^{\omega_2} (1 - \beta_{ij}^q)^{\Xi_i} \right)^{\eta_j}}{\prod_{j=1}^{\omega_1+1} \left(\prod_{i=1}^{\omega_2} (1 + \beta_{ij}^q)^{\Xi_i} \right)^{\eta_j} + \prod_{j=1}^{\omega_1+1} \left(\prod_{i=1}^{\omega_2} (1 - \beta_{ij}^q)^{\Xi_i} \right)^{\eta_j}}} \right\rangle$$

$$\otimes_{j=1}^{\omega_1+1} \left(\otimes_{i=1}^{\omega_2+1} (Q_{e_{ij}})^{\Xi_i} \right)^{\eta_j} = \left\langle \sqrt[q]{\frac{2 \prod_{j=1}^{\omega_1+1} \left(\prod_{i=1}^{\omega_2+1} (a_{ij}^q)^{\Xi_i} \right)^{\eta_j}}{\prod_{j=1}^{\omega_1+1} \left(\prod_{i=1}^{\omega_2+1} (2 - a_{ij}^q)^{\Xi_i} \right)^{\eta_j} + \prod_{j=1}^{\omega_1+1} \left(\prod_{i=1}^{\omega_2+1} (a_{ij}^q)^{\Xi_i} \right)^{\eta_j}}}, \sqrt[q]{\frac{\prod_{j=1}^{\omega_1+1} \left(\prod_{i=1}^{\omega_2+1} (1 + \beta_{ij}^q)^{\Xi_i} \right)^{\eta_j} - \prod_{j=1}^{\omega_1+1} \left(\prod_{i=1}^{\omega_2+1} (1 - \beta_{ij}^q)^{\Xi_i} \right)^{\eta_j}}{\prod_{j=1}^{\omega_1+1} \left(\prod_{i=1}^{\omega_2+1} (1 + \beta_{ij}^q)^{\Xi_i} \right)^{\eta_j} + \prod_{j=1}^{\omega_1+1} \left(\prod_{i=1}^{\omega_2+1} (1 - \beta_{ij}^q)^{\Xi_i} \right)^{\eta_j}}} \right\rangle$$

Now, for $m = \omega_1 + 1$ and $n = \omega_2 + 1$

$$\otimes_{j=1}^{\omega_1+1} \left(\otimes_{i=1}^{\omega_2+1} (Q_{e_{ij}})^{\Xi_i} \right)^{\eta_j} = \otimes_{j=1}^{\omega_1+1} \left(\otimes_{i=1}^{\omega_2} Q_{e_{ij}}^{\Xi_i} \otimes Q_{e_{(\omega_2+1)j}}^{\Xi_{\omega_2+1}} \right)^{\eta_j}$$

$$= \left(\otimes_{j=1}^{\omega_1+1} \otimes_{i=1}^{\omega_2} Q_{e_{ij}}^{\Xi_i} \right) \left(\otimes_{j=1}^{\omega_1+1} Q_{e_{(\omega_2+1)j}}^{\Xi_{\omega_2+1}} \right)^{\eta_j}$$

$$\left\langle \sqrt[q]{\frac{2 \prod_{j=1}^{\omega_1+1} \left(\prod_{i=1}^{\omega_2} (a_{ij}^q)^{\Xi_i} \right)^{\eta_j}}{\prod_{j=1}^{\omega_1+1} \left(\prod_{i=1}^{\omega_2} (2 - a_{ij}^q)^{\Xi_i} \right)^{\eta_j} + \prod_{j=1}^{\omega_1+1} \left(\prod_{i=1}^{\omega_2} (a_{ij}^q)^{\Xi_i} \right)^{\eta_j}}}, \sqrt[q]{\frac{\prod_{j=1}^{\omega_1+1} \left(\prod_{i=1}^{\omega_2} (1 + \beta_{ij}^q)^{\Xi_i} \right)^{\eta_j} - \prod_{j=1}^{\omega_1+1} \left(\prod_{i=1}^{\omega_2} (1 - \beta_{ij}^q)^{\Xi_i} \right)^{\eta_j}}{\prod_{j=1}^{\omega_1+1} \left(\prod_{i=1}^{\omega_2} (1 + \beta_{ij}^q)^{\Xi_i} \right)^{\eta_j} + \prod_{j=1}^{\omega_1+1} \left(\prod_{i=1}^{\omega_2} (1 - \beta_{ij}^q)^{\Xi_i} \right)^{\eta_j}}} \right\rangle \otimes \left\langle \sqrt[q]{\frac{2 \prod_{j=1}^{\omega_1+1} \left((a_{(\omega_2+1)j}^q)^{\Xi_{\omega_2+1}} \right)^{\eta_j}}{\prod_{j=1}^{\omega_1+1} \left((2 - a_{(\omega_2+1)j}^q)^{\Xi_{\omega_2+1}} \right)^{\eta_j} + \prod_{j=1}^{\omega_1+1} \left((a_{(\omega_2+1)j}^q)^{\Xi_{\omega_2+1}} \right)^{\eta_j}}}, \sqrt[q]{\frac{\prod_{j=1}^{\omega_1+1} \left((1 + \beta_{(\omega_2+1)j}^q)^{\Xi_{\omega_2+1}} \right)^{\eta_j} - \prod_{j=1}^{\omega_1+1} \left((1 - \beta_{(\omega_2+1)j}^q)^{\Xi_{\omega_2+1}} \right)^{\eta_j}}{\prod_{j=1}^{\omega_1+1} \left((1 + \beta_{(\omega_2+1)j}^q)^{\Xi_{\omega_2+1}} \right)^{\eta_j} + \prod_{j=1}^{\omega_1+1} \left((1 - \beta_{(\omega_2+1)j}^q)^{\Xi_{\omega_2+1}} \right)^{\eta_j}}} \right\rangle$$

$$= \left\langle \sqrt[q]{\frac{2 \prod_{j=1}^{\omega_1+1} \left(\prod_{i=1}^{\omega_2+1} (a_{ij}^q)^{\Xi_i} \right)^{\eta_j}}{\prod_{j=1}^{\omega_1+1} \left(\prod_{i=1}^{\omega_2+1} (2 - a_{ij}^q)^{\Xi_i} \right)^{\eta_j} + \prod_{j=1}^{\omega_1+1} \left(\prod_{i=1}^{\omega_2+1} (a_{ij}^q)^{\Xi_i} \right)^{\eta_j}}}, \sqrt[q]{\frac{\prod_{j=1}^{\omega_1+1} \left(\prod_{i=1}^{\omega_2+1} (1 + \beta_{ij}^q)^{\Xi_i} \right)^{\eta_j} - \prod_{j=1}^{\omega_1+1} \left(\prod_{i=1}^{\omega_2+1} (1 - \beta_{ij}^q)^{\Xi_i} \right)^{\eta_j}}{\prod_{j=1}^{\omega_1+1} \left(\prod_{i=1}^{\omega_2+1} (1 + \beta_{ij}^q)^{\Xi_i} \right)^{\eta_j} + \prod_{j=1}^{\omega_1+1} \left(\prod_{i=1}^{\omega_2+1} (1 - \beta_{ij}^q)^{\Xi_i} \right)^{\eta_j}}} \right\rangle$$

$$= \otimes_{j=1}^{\omega_1+1} \left(\otimes_{i=1}^{\omega_2+1} (Q_{e_{ij}})^{\Xi_i} \right)^{\eta_j}$$

Therefore, it holds for $m = \omega_1 + 1$ and $n = \omega_2 + 1$.

Proposition 1: If $Q_{e_{ij}} = (\alpha_{ij}, \beta_{ij})$ be a collection of Lq-ROFHSSs where $i, j = 1, 2, \dots, n, m$ and where $\Xi_i > 0, \eta_j > 0$ denotes the weights of experts and attributes, and $\sum_{i=1}^n \Xi_i = \sum_{j=1}^m \eta_j = 1$.

Idempotency: If $Q_{ij} = Q = (\alpha_{ij}, \beta_{ij})$ be the collection of Lq-ROFHSSs such that $Q_{e_{ij}} = (\alpha_{ij}, \beta_{ij}) = (\mu, \beta) = Q_e \forall (i, 1, \dots, n)$ and $(j = 1, \dots, m)$. Then,

$$Lq - \text{ROFHSSs}(Q_{e_{11}}, Q_{e_{12}}, \dots, Q_{e_{nm}}) = (\mu, \beta) = Q_e$$

Proof: Since $Q_{e_{ij}} = (\alpha_{ij}, \beta_{ij}) = Q_e, \forall (i = 1, \dots, n, j = 1, \dots, m)$ so:

$$\text{Lq-ROFHSSs}(Q_{e_{11}}, Q_{e_{12}}, \dots, Q_{e_{nm}}) = \left\langle \sqrt[q]{\frac{2 \prod_{j=1}^m \left(\prod_{i=1}^n (a_{ij}^q)^{\Xi_i} \right)^{\eta_j}}{\prod_{j=1}^m \left(\prod_{i=1}^n (2 - a_{ij}^q)^{\Xi_i} \right)^{\eta_j} + \prod_{j=1}^m \left(\prod_{i=1}^n (a_{ij}^q)^{\Xi_i} \right)^{\eta_j}}}, \sqrt[q]{\frac{\prod_{j=1}^m \left(\prod_{i=1}^n (1 + \beta_{ij}^q)^{\Xi_i} \right)^{\eta_j} - \prod_{j=1}^m \left(\prod_{i=1}^n (1 - \beta_{ij}^q)^{\Xi_i} \right)^{\eta_j}}{\prod_{j=1}^m \left(\prod_{i=1}^n (1 + \beta_{ij}^q)^{\Xi_i} \right)^{\eta_j} + \prod_{j=1}^m \left(\prod_{i=1}^n (1 - \beta_{ij}^q)^{\Xi_i} \right)^{\eta_j}}} \right\rangle$$

$$= \left\langle \sqrt[q]{\frac{2 \left((a_{ij}^q)^{\sum_{i=1}^n \Xi_i} \right)^{\sum_{j=1}^m \eta_j}}{\left((2 - a_{ij}^q)^{\sum_{i=1}^n \Xi_i} \right)^{\sum_{j=1}^m \eta_j} + \left((a_{ij}^q)^{\sum_{i=1}^n \Xi_i} \right)^{\sum_{j=1}^m \eta_j}}}, \sqrt[q]{\frac{\left((1 + \beta_{ij}^q)^{\sum_{i=1}^n \Xi_i} \right)^{\sum_{j=1}^m \eta_j} - \left((1 - \beta_{ij}^q)^{\sum_{i=1}^n \Xi_i} \right)^{\sum_{j=1}^m \eta_j}}{\left((1 + \beta_{ij}^q)^{\sum_{i=1}^n \Xi_i} \right)^{\sum_{j=1}^m \eta_j} + \left((1 - \beta_{ij}^q)^{\sum_{i=1}^n \Xi_i} \right)^{\sum_{j=1}^m \eta_j}}} \right\rangle$$

$$= \left\langle \sqrt[q]{\frac{2a_{ij}^q}{(2 - a_{ij}^q) + a_{ij}^q}}, \sqrt[q]{\frac{(1 + \beta_{ij}^q) - (1 - \beta_{ij}^q)}{(1 + \beta_{ij}^q) + (1 - \beta_{ij}^q)}} \right\rangle = (\alpha_{ij}, \beta_{ij}) = Q_e$$

Boundedness: If $Q_{ij} = (\alpha_{ij}, \beta_{ij})$ be a collection of Lq-ROFHSSs and $Q_{\min} = \min(Q_{ij}), Q_{\max} = \max(Q_{ij})$. Then,

$$Q_{\min} \leq Lq - \text{ROFHSSs}(Q_{e_{11}}, Q_{e_{12}}, \dots, Q_{e_{nm}}) \leq Q_{\max}$$

Proof: Let $f(x) = \sqrt[q]{\frac{2-x^q}{x^q}}$, serves as a function defined for $x \in [0, 1]$, then

$$\frac{d}{dx}(f(x)) = -\frac{\left[\frac{q}{x} + \frac{q}{x^{1+q}}(2 - x^q)\right] \left[\frac{1}{\theta^q}(2 - x^q)^{-1+\frac{1}{q}}\right]}{q} \cdot \frac{-2}{x^3} < 0.$$

So, this shows that $f(x)$ represents decreasing function on $[0, 1]$. Since $\alpha_{\min} \leq \alpha_{ij} \leq \alpha_{\max} \forall i, j$,

$$f(\alpha_{\max}) \leq f(\alpha_{ij}) \leq f(\alpha_{\min}).$$

So we have

$$\sqrt[q]{\frac{2 - \alpha_{\max}^q}{\alpha_{\max}^q}} \leq \sqrt[q]{\frac{2 - \alpha_{ij}^q}{\alpha_{ij}^q}} \leq \sqrt[q]{\frac{2 - \alpha_{\min}^q}{\alpha_{\min}^q}}, \quad \forall (i = 1, 2, \dots, n), (j = 1, 2, \dots, m).$$

Where Ξ_i and η_j represents the weight vectors for experts and parameters, correspondingly, such as $\Xi_i > 0$, $\sum_{i=1}^n \Xi_i = 1$, $\eta_j > 0$, $\sum_{j=1}^m \eta_j = 1$.

Also $g(y) = \sqrt[q]{\frac{1-y^q}{1+y^q}}$ be a function defined as $y \in [0, 1]$ then $\frac{d}{dy}(g(y)) = -\frac{\left[\frac{q}{y} + \frac{q}{y^{1+q}}(2-y^q)\right] \left[\frac{1}{\theta^q}(2-y^q)^{-1+\frac{1}{q}}\right]}{q} \frac{-2}{y^3} < 0$, which shows that the function $g(y)$ is decreasing. Hence $\beta_{\min} \leq \beta_{ij} \leq \beta_{\max}$ for $g(\beta_{\max}) \leq g(\beta_{ij}) \leq g(\beta_{\min})$ given that the function is decreasing. Thus, we have $\sqrt[q]{\frac{1-\beta_{\max}^q}{1+\beta_{\max}^q}} \leq \sqrt[q]{\frac{1-\beta_{ij}^q}{1+\beta_{ij}^q}} \leq \sqrt[q]{\frac{1-\beta_{\min}^q}{1+\beta_{\min}^q}}$, $\forall (i = 1, 2, \dots, n)$ and $(j = 1, 2, \dots, m)$, where Ξ_i and η_j denotes the weight vectors for experts and parameters, correspondingly, such as $\Xi_i > 0$, $\sum_{i=1}^n \Xi_i = 1$, $\eta_j > 0$, $\sum_{j=1}^m \eta_j = 1$.

Since,

$$\begin{aligned} Q_{\min} &\leq Lq - \text{ROFHSs}(Q_{e_{11}}, Q_{e_{12}}, \dots, Q_{e_{nm}}) \leq Q_{\max} \\ &\Leftrightarrow \sqrt[q]{\prod_{j=1}^m \left(\prod_{i=1}^n \left(\frac{2 - \alpha_{\max}^q}{\alpha_{\max}^q} \right)^{\Xi_i} \right)^{\eta_j}} \leq \sqrt[q]{\prod_{j=1}^m \left(\prod_{i=1}^n \left(\frac{2 - \alpha_{ij}^q}{\alpha_{ij}^q} \right)^{\Xi_i} \right)^{\eta_j}} \leq \sqrt[q]{\prod_{j=1}^m \left(\prod_{i=1}^n \left(\frac{2 - \alpha_{\min}^q}{\alpha_{\min}^q} \right)^{\Xi_i} \right)^{\eta_j}} \\ &\Leftrightarrow \sqrt[q]{\left(\left(\frac{2 - \alpha_{\max}^q}{\alpha_{\max}^q} \right)^{\sum_{i=1}^n \Xi_i} \right)^{\sum_{j=1}^m \eta_j}} \leq \sqrt[q]{\prod_{j=1}^m \left(\prod_{i=1}^n \left(\frac{2 - \alpha_{ij}^q}{\alpha_{ij}^q} \right)^{\Xi_i} \right)^{\eta_j}} \leq \sqrt[q]{\left(\left(\frac{2 - \alpha_{\min}^q}{\alpha_{\min}^q} \right)^{\sum_{i=1}^n \Xi_i} \right)^{\sum_{j=1}^m \eta_j}} \\ &\Leftrightarrow \sqrt[q]{\left(\frac{2 - \alpha_{\max}^q}{\alpha_{\max}^q} \right)} \leq \sqrt[q]{\prod_{j=1}^m \left(\prod_{i=1}^n \left(\frac{2 - \alpha_{ij}^q}{\alpha_{ij}^q} \right)^{\Xi_i} \right)^{\eta_j}} \leq \sqrt[q]{\left(\frac{2 - \alpha_{\min}^q}{\alpha_{\min}^q} \right)} \\ &\Leftrightarrow \sqrt[q]{1 + \left(\frac{2 - \alpha_{\max}^q}{\alpha_{\max}^q} \right)} \leq \sqrt[q]{1 + \prod_{j=1}^m \left(\prod_{i=1}^n \left(\frac{2 - \alpha_{ij}^q}{\alpha_{ij}^q} \right)^{\Xi_i} \right)^{\eta_j}} \leq \sqrt[q]{1 + \left(\frac{2 - \alpha_{\min}^q}{\alpha_{\min}^q} \right)} \\ &\Leftrightarrow \sqrt[q]{\frac{2}{\alpha_{\max}^q}} \leq \sqrt[q]{1 + \prod_{j=1}^m \left(\prod_{i=1}^n \left(\frac{2 - \alpha_{ij}^q}{\alpha_{ij}^q} \right)^{\Xi_i} \right)^{\eta_j}} \leq \sqrt[q]{\frac{2}{\alpha_{\min}^q}} \\ &\Leftrightarrow \sqrt[q]{\frac{\alpha_{\min}^q}{2}} \leq \frac{1}{\sqrt[q]{1 + \prod_{j=1}^m \left(\prod_{i=1}^n \left(\frac{2 - \alpha_{ij}^q}{\alpha_{ij}^q} \right)^{\Xi_i} \right)^{\eta_j}}} \leq \sqrt[q]{\frac{\alpha_{\max}^q}{2}} \Leftrightarrow \alpha_{\min} \leq q \sqrt[q]{\frac{2}{1 + \prod_{j=1}^m \left(\prod_{i=1}^n \left(\frac{2 - \alpha_{ij}^q}{\alpha_{ij}^q} \right)^{\Xi_i} \right)^{\eta_j}}} \leq \alpha_{\max} \\ &\Leftrightarrow \alpha_{\min} \leq \frac{\sqrt[q]{2 \prod_{j=1}^m \left(\prod_{i=1}^n \left(\alpha_{ij}^q \right)^{\Xi_i} \right)^{\eta_j}}}{\sqrt[q]{\prod_{j=1}^m \left(\prod_{i=1}^n \left(2 - \alpha_{ij}^q \right)^{\Xi_i} \right)^{\eta_j} + \prod_{j=1}^m \left(\prod_{i=1}^n \left(\alpha_{ij}^q \right)^{\Xi_i} \right)^{\eta_j}}} \leq \alpha_{\max} \end{aligned} \tag{12}$$

Let $g(y) = \sqrt[q]{\frac{1-y^q}{1+y^q}}$, $y \in [0, 1]$. Then, $\frac{d}{dy}(g(y)) = -\frac{\left[\frac{q}{y} + \frac{q}{y^{1+q}}(2-y^q)\right] \left[\frac{1}{\theta^q}(2-y^q)^{-1+\frac{1}{q}}\right]}{q} \frac{-2}{y^3} < 0$. So, $g(y)$ is decreasing function on $[0, 1]$. Thus, $\beta_{\min} \leq \beta_{ij} \leq \beta_{\max} \forall i, j$. So, $g(\beta_{\max}) \leq g(\beta_{ij}) \leq g(\beta_{\min}) \forall i, j \Rightarrow \sqrt[q]{\frac{1-\beta_{\max}^q}{1+\beta_{\max}^q}} \leq \sqrt[q]{\frac{1-\beta_{ij}^q}{1+\beta_{ij}^q}} \leq \sqrt[q]{\frac{1-\beta_{\min}^q}{1+\beta_{\min}^q}}$, Ξ_i and η_j shows the weight vectors for experts and parameters, correspondingly, like $\Xi_i > 0$, $\sum_{i=1}^n \Xi_i = 1$, $\eta_j > 0$, $\sum_{j=1}^m \eta_j = 1$.

$$\begin{aligned}
&\Rightarrow \sqrt[q]{\prod_{j=1}^m \left(\prod_{i=1}^n \left(\frac{1 - \beta_{\max}^q}{1 + \beta_{\max}^q} \right)^{\Xi_i} \right)^{\eta_j}} \leq \sqrt[q]{\prod_{j=1}^m \left(\prod_{i=1}^n \left(\frac{1 - \beta_{ij}^q}{1 + \beta_{ij}^q} \right)^{\Xi_i} \right)^{\eta_j}} \leq \sqrt[q]{\prod_{j=1}^m \left(\prod_{i=1}^n \left(\frac{1 - \beta_{\min}^q}{1 + \beta_{\min}^q} \right)^{\Xi_i} \right)^{\eta_j}} \\
&\Leftrightarrow \sqrt[q]{\left(\left(\frac{1 - \beta_{\max}^q}{1 + \beta_{\max}^q} \right)^{\sum_{i=1}^n \Xi_i} \right)^{\sum_{j=1}^m \eta_j}} \leq \sqrt[q]{\prod_{j=1}^m \left(\prod_{i=1}^n \left(\frac{1 - \beta_{ij}^q}{1 + \beta_{ij}^q} \right)^{\Xi_i} \right)^{\eta_j}} \leq \sqrt[q]{\left(\left(\frac{1 - \beta_{\min}^q}{1 + \beta_{\min}^q} \right)^{\sum_{i=1}^n \Xi_i} \right)^{\sum_{j=1}^m \eta_j}} \\
&\Leftrightarrow \sqrt[q]{\left(\frac{1 - \beta_{\max}^q}{1 + \beta_{\max}^q} \right)} \leq \sqrt[q]{\prod_{j=1}^m \left(\prod_{i=1}^n \left(\frac{1 - \beta_{ij}^q}{1 + \beta_{ij}^q} \right)^{\Xi_i} \right)^{\eta_j}} \leq \sqrt[q]{\left(\frac{1 - \beta_{\min}^q}{1 + \beta_{\min}^q} \right)} \\
&\Leftrightarrow \sqrt[q]{1 + \left(\frac{1 - \beta_{\max}^q}{1 + \beta_{\max}^q} \right)} \leq \sqrt[q]{1 + \prod_{j=1}^m \left(\prod_{i=1}^n \left(\frac{1 - \beta_{ij}^q}{1 + \beta_{ij}^q} \right)^{\Xi_i} \right)^{\eta_j}} \leq \sqrt[q]{1 + \left(\frac{1 - \beta_{\min}^q}{1 + \beta_{\min}^q} \right)} \\
&\Leftrightarrow \sqrt[q]{\frac{2}{1 + \beta_{\max}^q}} \leq \sqrt[q]{1 + \prod_{j=1}^m \left(\prod_{i=1}^n \left(\frac{1 - \beta_{ij}^q}{1 + \beta_{ij}^q} \right)^{\Xi_i} \right)^{\eta_j}} \leq \sqrt[q]{\frac{2}{1 + \beta_{\min}^q}} \\
&\Leftrightarrow \sqrt[q]{\frac{1 + \beta_{\min}^q}{2}} \leq \frac{1}{\sqrt[q]{1 + \prod_{j=1}^m \left(\prod_{i=1}^n \left(\frac{1 - \beta_{ij}^q}{1 + \beta_{ij}^q} \right)^{\Xi_i} \right)^{\eta_j}}} \leq \sqrt[q]{\frac{1 + \beta_{\max}^q}{2}} \\
&\Leftrightarrow \sqrt[q]{1 + \beta_{\min}^q} \leq \sqrt[q]{\frac{2}{1 + \prod_{j=1}^m \left(\prod_{i=1}^n \left(\frac{1 - \beta_{ij}^q}{1 + \beta_{ij}^q} \right)^{\Xi_i} \right)^{\eta_j}}} \leq \sqrt[q]{1 + \beta_{\max}^q} \Leftrightarrow \beta_{\min} \leq q \sqrt[q]{\frac{2}{1 + \prod_{j=1}^m \left(\prod_{i=1}^n \left(\frac{1 - \beta_{ij}^q}{1 + \beta_{ij}^q} \right)^{\Xi_i} \right)^{\eta_j}}} - 1 \leq \beta_{\max} \\
&\Leftrightarrow \beta_{\min} \leq \frac{\sqrt[q]{\prod_{j=1}^m \left(\prod_{i=1}^n (1 + \beta_{ij}^q)^{\Xi_i} \right)^{\eta_j}} - \sqrt[q]{\prod_{j=1}^m \left(\prod_{i=1}^n (1 - \beta_{ij}^q)^{\Xi_i} \right)^{\eta_j}}}{\sqrt[q]{\prod_{j=1}^m \left(\prod_{i=1}^n (1 + \beta_{ij}^q)^{\Xi_i} \right)^{\eta_j}} + \sqrt[q]{\prod_{j=1}^m \left(\prod_{i=1}^n (1 - \beta_{ij}^q)^{\Xi_i} \right)^{\eta_j}}} \leq \beta_{\max}
\end{aligned}$$

(13)

Let $\omega = \text{Lq-ROFHSs } (Q_{e_{11}}, Q_{e_{12}}, \dots, Q_{e_{nm}}) = \langle \alpha_\omega, \beta_\omega \rangle = Q_\omega$, then employing Eq. (1).

$$\begin{aligned}
S(\omega) &= \alpha_\omega^q - \beta_\omega^q + \left(\frac{e^{\alpha_\omega q - \beta_\omega^q}}{e^{\alpha_\omega^q - \beta_\omega^q} + 1} - \frac{1}{2} \right) \pi_\omega^q \leq \left(\max_j \max_i \{ \alpha_{ij} \} \right)^q - \left(\min_j \min_i \{ \beta_{ij} \} \right)^q + \left(\frac{e^{(\max_j \max_i \{ \alpha_{ij} \})^q - (\min_j \min_i \{ \beta_{ij} \})^q}}{e^{(\max_j \max_i \{ \alpha_{ij} \})^q - (\min_j \min_i \{ \beta_{ij} \})^q} + 1} - \frac{1}{2} \right) \pi_{Q_{e_{ij}}}^q = S(Q_{\max}) \\
&\Rightarrow S(\omega) \leq S(Q_{\max}) \\
S(\omega) &= \alpha_\omega^q - \beta_\omega^q + \left(\frac{e^{\alpha_\omega q - \beta_\omega^q}}{e^{\alpha_\omega^q - \beta_\omega^q} + 1} - \frac{1}{2} \right) \pi_\omega^q \geq \left(\min_j \min_i \{ \alpha_{ij} \} \right)^q - \left(\max_j \max_i \{ \beta_{ij} \} \right)^q + \left(\frac{e^{(\min_j \min_i \{ \alpha_{ij} \})^q - (\max_j \max_i \{ \beta_{ij} \})^q}}{e^{(\min_j \min_i \{ \alpha_{ij} \})^q - (\max_j \max_i \{ \beta_{ij} \})^q} + 1} - \frac{1}{2} \right) \pi_{Q_{e_{ij}}}^q = S(Q_{\min}) \\
&\Rightarrow S(\omega) \geq S(Q_{\min})
\end{aligned}$$

4 Extended Technique for Order Preference by Similarity to Ideal Solution Method in Linguistic q-Rung Orthopair Fuzzy Hypersoft Sets

This section presents the MAGDM approach known as TOPSIS based on the proposed weighted correlation coefficient of L_q -ROFHSs. Figures 1 and 2 illustrate the overall workflow and computational procedure of the proposed method, respectively.

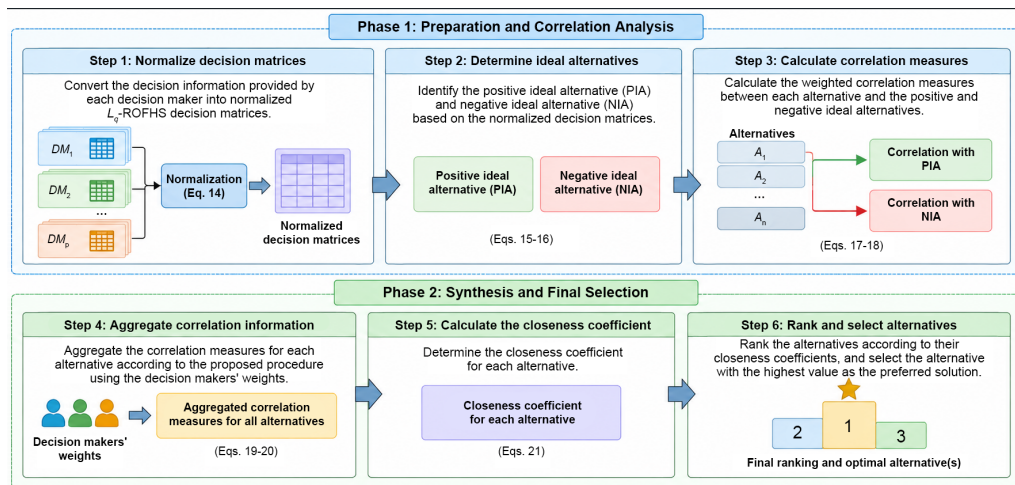


Figure 1. Flowchart of the proposed linguistic q-rung orthopair fuzzy hypersoft sets (Lq-ROFHS)-based technique for order preference by similarity to ideal solution (TOPSIS) framework

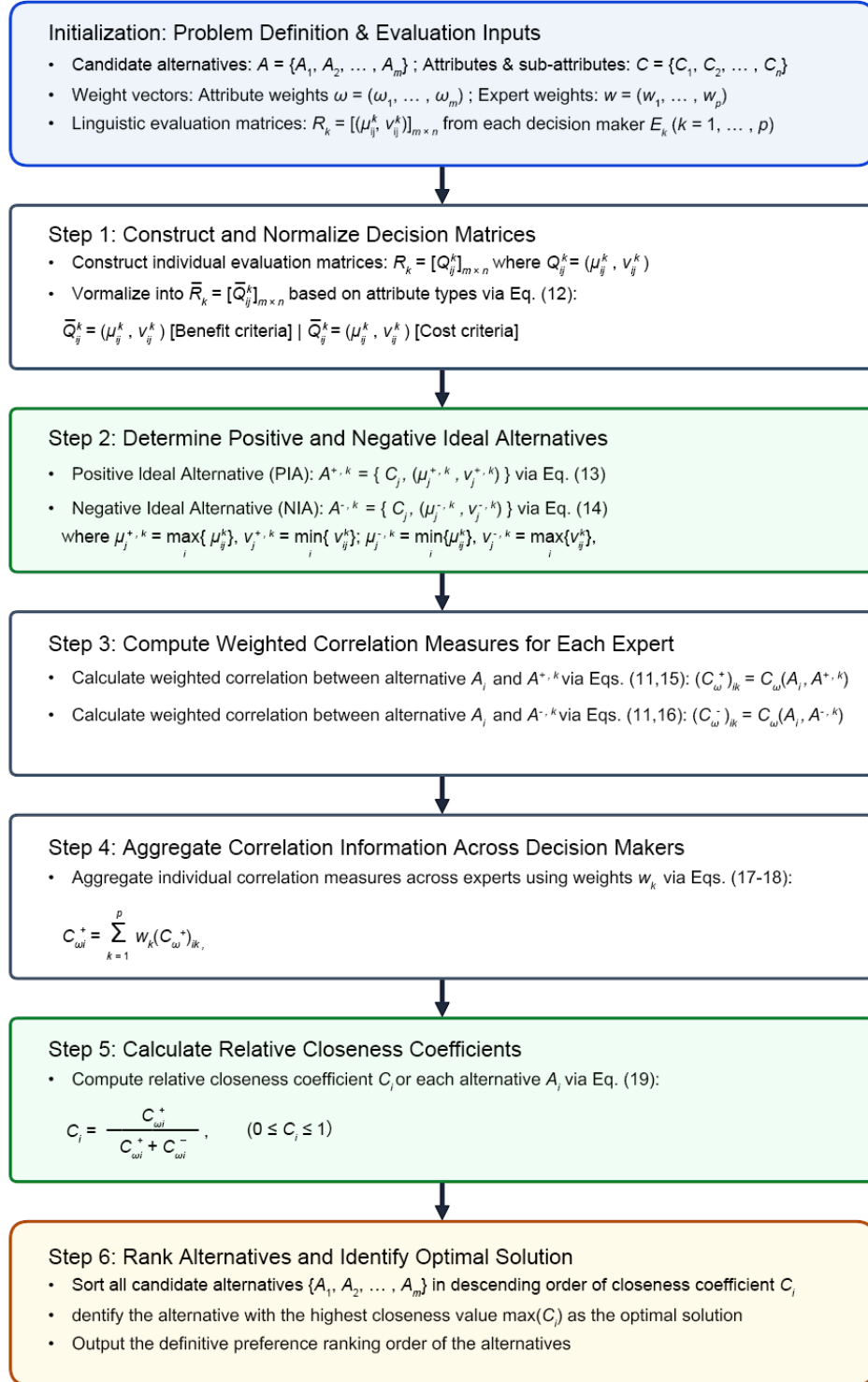


Figure 2. Technique for order preference by similarity to ideal solution algorithm based on the proposed correlation measures under the linguistic q-rung orthopair fuzzy hypersoft sets (Lq-ROFHS) environment

Let $\{\mathfrak{S}_1, \mathfrak{S}_2, \dots, \mathfrak{S}_j\}$ and $x^i = \{x^p, x^q, \dots, x^k\}$ Alternatives and attributes, respectively. Also, let $\omega_1, \omega_2, \dots, \omega_n$ be the n weights of the element x^i , such that $x^i \geq 1, \omega_i \geq 1$ with $\sum_{i=1}^n \omega_i = 1$. Now let $\mathfrak{d} = \{\mathfrak{d}_1, \mathfrak{d}_2, \dots, \mathfrak{d}_n\}$ be a set of DM experts $\mathcal{DM}_e$ with weight $w_1, w_2, \dots, w_n$ be the n weights of the element $x^i \geq 1$, such that $\mathfrak{d} \geq 1, w_i \geq 1$, with $\sum_{i=1}^n w_i = 1$. Every $\mathfrak{d}_i$ evaluates the attributes x^i of the alternatives $\mathfrak{S}_i$ by utilizing Lq-ROFNs in the form of $\partial_i = (\ell_{\mu^l(\mathfrak{S})}(x^l), \ell_{v^l(\mathfrak{S})}(x^l))$ to construct a decision matrix $\bar{\mathfrak{R}}^i = [\bar{\mathfrak{Q}}_{jk}^i]_{n \times n}$ for each $\mathcal{DM}_e \mathfrak{d}_i$. And

can be represented as:

$$\overline{\mathfrak{R}}^i = \begin{matrix} \mathfrak{T}_i \\ \vdots \\ \mathfrak{T}_n \end{matrix} \begin{bmatrix} x^p & x^q & \cdots & x^r \\ x_1^p & x_1^q & \cdots & x_1^r \\ \vdots & \vdots & \ddots & \vdots \\ x_n^p & x_n^q & \cdots & x_n^r \end{bmatrix}$$

Step 1: Convert the $\overline{\mathfrak{R}}^i = [\mathfrak{Q}_{jk}^i]_{n \times n}$ of each $\mathcal{DM}_e \mathfrak{D}_i = (\overline{\ell_{\mu^i(\mathfrak{S})}(x^i)}, \overline{\ell_{\nu^i(\mathfrak{S})}(x^i)})$ into the normalized decision matrix $\mathcal{NDM}_e \mathfrak{R}^i = [\mathfrak{Q}_{jk}^i]_{n \times n}$ for each $\mathcal{DM}_e \mathfrak{D}_i = (\ell_{\mu^i(\mathfrak{S})}(x^i), \ell_{\nu^i(\mathfrak{S})}(x^i))$.

$\overline{\mathfrak{R}}^1 = [\mathfrak{Q}_{jk}^1]_{n \times n} = (\overline{\ell_{\mu^1(\mathfrak{S})}(x^1)}, \overline{\ell_{\nu^1(\mathfrak{S})}(x^1)})_{n \times n}$, $\overline{\mathfrak{R}}^2 = [\mathfrak{Q}_{jk}^2]_{n \times n} = (\overline{\ell_{\mu^2(\mathfrak{S})}(x^2)}, \overline{\ell_{\nu^2(\mathfrak{S})}(x^2)})_{n \times n}$, ..., $\overline{\mathfrak{R}}^n = [\mathfrak{Q}_{jk}^n]_{n \times n} = (\overline{\ell_{\mu^n(\mathfrak{S})}(x^n)}, \overline{\ell_{\nu^n(\mathfrak{S})}(x^n)})_{n \times n}$ into a normalized decision matrix $\mathcal{NDM}_e$.

$\mathfrak{R}^1 = [\mathfrak{Q}_{jk}^1]_{n \times n} = (\ell_{\mu^1(\mathfrak{S})}(x^i), \ell_{\nu^1(\mathfrak{S})}(x^i))_{n \times n}$, $\mathfrak{R}^2 = [\mathfrak{Q}_{jk}^2]_{n \times n} = (\ell_{\mu^2(\mathfrak{S})}(x^i), \ell_{\nu^2(\mathfrak{S})}(x^i))_{n \times n}$, ..., $\mathfrak{R}^n = [\mathfrak{Q}_{jk}^n]_{n \times n} = (\ell_{\mu^n(\mathfrak{S})}(x^i), \ell_{\nu^n(\mathfrak{S})}(x^i))_{n \times n}$ represented below with Eq. (14);

$$\mathfrak{Q}_{jk}^i = \begin{cases} (\ell_{\mu^i(\mathfrak{S})}(x^i), \ell_{\nu^i(\mathfrak{S})}(x^i)) & | \text{beneficial attribute} \\ (\ell_{\nu^i(\mathfrak{S})}(x^i), \ell_{\mu^i(\mathfrak{S})}(x^i)) & | \text{cost attribute} \end{cases}, \text{ where; } i, j, k = 1, 2, \dots, n \quad (14)$$

Step 2: Now we need to find the positive ideal alternative $(\mathcal{P}_{i\mathcal{A}})(\mathfrak{S}^+)^i$ and the negative ideal alternative $(\mathcal{N}_{i\mathcal{A}})(\mathfrak{S}^-)^i$ using the following Eqs. (15) and (16).

$$(\mathfrak{S}^+)^i = \langle x^i, \ell_{(\mu^i)^+}, \ell_{(\nu^i)^+} \rangle \quad (15)$$

$$(\mathfrak{S}^-)^i = \langle x^i, \ell_{(\nu^i)^-}, \ell_{(\mu^i)^-} \rangle \quad (16)$$

where,

$$\begin{aligned} (\mu^i)^+ &= \max_{\delta^i} \text{ from } \{\mathfrak{Q}_{jk}^i\}, (\mu^i)^- = \min_{\delta^i} \text{ from } \{\mathfrak{Q}_{jk}^i\}, (\nu^i)^+ = \min_{\delta^i} \text{ from } \{\mathfrak{Q}_{jk}^i\}, \\ (\nu^i)^- &= \max_{\delta^i} \text{ from } \{\mathfrak{Q}_{jk}^i\} \text{ and for the degree of hesitancy } (\xi^i)^+ = \left| (t^q - (\mu^i)^{+q} - (\nu^i)^{+q}) \right|^{\frac{1}{q}}, \\ (\xi^i)^- &= \left| (t^q - (\mu^i)^{-q} - (\nu^i)^{-q}) \right|^{\frac{1}{q}} \text{ for } q \geq 1. \end{aligned}$$

Step 3: Now, using Eq. (11), obtain $(\mathfrak{C}_\omega^+)^i$ between $\mathfrak{S}_i$ and $(\mathcal{P}_{i\mathcal{A}})(\mathfrak{S}^+)^i$. Similarly, obtain $(\mathfrak{C}_\omega^-)^i$ between $\mathfrak{S}_i$ and $(\mathcal{N}_{i\mathcal{A}})(\mathfrak{S}^-)^i$ for each $\mathcal{DM}_e = \mathfrak{D}_i$ using the Eqs. (17) and (18). Where ω_i is the weight of the element x^i , and $\omega_i \geq 1$ with $\sum_{i=1}^n \omega_i = 1$.

$$(\mathfrak{C}_\omega^+)^i = \mathfrak{C}_\omega \left(\mathfrak{S}_i, (\mathfrak{S}^+)^i \right) \quad (17)$$

$$(\mathfrak{C}_\omega^-)^i = \mathfrak{C}_\omega \left(\mathfrak{S}_i, (\mathfrak{S}^-)^i \right) \quad (18)$$

Step 4: Calculate aggregated $\mathfrak{C}_\omega^+$ and $\mathfrak{C}_\omega^-$ for the alternative $\mathfrak{J}_i$, using the following Eqs. (19) and (20).

$$\mathfrak{C}_\omega^+ = \omega_i (\mathfrak{C}_\omega^+)^i \quad (19)$$

$$\mathfrak{C}_\omega^- = \omega_i (\mathfrak{C}_\omega^-)^i \quad (20)$$

Step 5: Calculate the coefficient of closeness $\mathbb{C}_i$ of the alternative $\mathfrak{J}_i$ based on $\mathfrak{C}_\omega^+$ and $\mathfrak{C}_\omega^-$ using the Eq. (21).

$$\mathbb{C}_i = \frac{\mathfrak{C}_\omega^+}{\mathfrak{C}_\omega^+ + \mathfrak{C}_\omega^-} \quad (21)$$

Step 6: Based on $\mathbb{C}_i$ rank the alternatives $\mathfrak{S}_i$, the alternative with a higher value of $\mathbb{C}$ will be the optimal choice.

The overall workflow of the proposed Lq-ROFHS-based TOPSIS framework is illustrated in Figure 1. The procedure begins with the construction and normalization of the decision information, followed by the identification of the positive and negative ideal alternatives. The proposed correlation measures are then used to evaluate the relationships between each candidate alternative and the ideal alternatives. After aggregating the decision information, the closeness coefficients are calculated to obtain the final preference ranking.

The computational procedure of the proposed MAGDM framework is further summarized in Figure 2. The algorithm provides a step-by-step representation of the decision process, from the input of alternatives, criteria, and decision-maker assessments to the construction of the corresponding decision matrices and the determination of the positive and negative ideal alternatives. The proposed weighted correlation measures are subsequently applied, and the resulting information is used to calculate the closeness coefficient C_i for each alternative. The alternatives are finally ranked according to their C_i values to determine the preferred solution.

5 Practical Implications of the Proposed Technique for Order Preference by Similarity to Ideal Solution Model

The Lq-ROFHS framework and the proposed correlation measures provide a flexible analytical structure for decision problems involving multiple criteria, subdivided attributes, linguistic assessments, and uncertain information. By combining the representation capability of Lq-ROFHSs with correlation-based evaluation, the proposed framework can accommodate heterogeneous expert judgements while preserving the contextual information associated with individual attributes and their subdivisions. These characteristics make the approach applicable to a range of decision-support problems in which precise quantitative information is limited or difficult to obtain. The practical relevance of the proposed framework can be illustrated across several application domains.

Medical Diagnosis: In healthcare decision-making, symptoms and clinical indicators may differ in importance across patient characteristics or diagnostic subcategories, such as age groups and severity levels. The Lq-ROFHS framework can represent such subdivided evaluation information while accommodating linguistic and uncertain expert assessments. The proposed correlation measures can then support the comparison of diagnostic alternatives when symptom information or test outcomes cannot be expressed with complete precision.

Environmental Management: Sustainability-related environmental decisions often require alternatives to be evaluated against multiple and potentially conflicting criteria, including efficiency, cost, and environmental impact. For problems such as wastewater-treatment technology selection, the Lq-ROFHS framework can represent both the criteria and their relevant sub-attributes while retaining uncertainty in expert evaluations. The proposed correlation-based decision procedure can further support alternative comparison when environmental information or stakeholder preferences are expressed linguistically or contain uncertainty.

Social Sciences and Policy Making: Policy evaluation and prioritization frequently involve qualitative judgements and criteria whose relevance varies across regions, population groups, or implementation contexts. The Lq-ROFHS framework provides a structured means of representing such multidimensional information and can support the comparison of policy alternatives when assessments depend on heterogeneous stakeholder preferences and linguistic evaluations.

Engineering Design and Evaluation: Engineering and product-design decisions commonly involve several technical and non-technical criteria with different levels of importance and uncertainty. The proposed framework can represent subdivided design attributes and linguistic expert assessments within a common analytical structure. Its correlation-based evaluation mechanism can therefore support the comparison and prioritization of engineering alternatives when precise information is unavailable or when preferences are expressed qualitatively.

These application areas illustrate the broader relevance of Lq-ROFHS-based correlation measures to structured decision problems characterized by uncertain, linguistic, and multidimensional information. By integrating HSs, linguistic information, and q-rung orthopair fuzzy representation within a TOPSIS-based decision procedure, the proposed framework extends conventional alternative-ranking approaches to settings in which attributes require further subdivision and expert assessments cannot be represented adequately by precise numerical values. In the present study, this capability is examined through a sustainability-oriented application involving resilient microgrid selection for a rural clinic.

5.1 Numerical Example for Resilient Microgrid Selection for a Rural Clinic

Reliable electricity supply remains a critical requirement for rural healthcare facilities, where power interruptions can affect clinical services, vaccine storage, communication systems, and the operation of essential medical equipment. Selecting an appropriate microgrid configuration is therefore not solely an economic decision; it requires the simultaneous consideration of reliability, cost, environmental performance, social suitability, and other resilience-related factors. These criteria may conflict with one another, while expert assessments may contain uncertainty and may be expressed more naturally in linguistic rather than precise numerical terms.

The proposed Lq-ROFHS-based TOPSIS framework provides a structured approach for representing these characteristics within a unified decision process. Linguistic q-rung orthopair fuzzy hypersoft information is used to retain uncertainty, hesitation, and subdivided attribute information in the expert assessments, while the proposed correlation measures support the comparison of the candidate alternatives. The following numerical example applies the framework to the selection of a resilient microgrid configuration for a rural clinic.

Consider three alternatives:

A_1 : Solar PV + battery

A_2 : Solar PV + battery + backup diesel

A_3 : Diesel - only baseline

and 4 criteria with sub-divided attributes.

C_1 Cost efficiency: {CAPEX saving 0.40, OPEX saving 0.40, price stability 0.20}

C_2 Reliability: {uptime 0.50, supply resilience 0.30, maintenance simplicity 0.20}

C_3 Environmental benefit: {CO₂ reduction 0.60, noise reduction 0.20, no waste 0.20}

C_4 Scalability: {ease of expansion 0.40, local jobs 0.35, training support 0.25}

Three decision makers $DM_s = \{\mathfrak{d}_1, \mathfrak{d}_2, \mathfrak{d}_3\}$ will assign the weights $w_1 = 0.40$, $w_2 = 0.35$, and $w_3 = 0.25$. The evaluation uses the Lq-ROFHSs framework with $q = 3$, and the extended TOPSIS based on the proposed (weighted) correlation coefficient as in Section 3 and Section 4. Every $\mathfrak{d}_i$ evaluates the attributes x^i of the alternatives $\mathfrak{S}_i$ by utilizing Lq-ROFNs in the form of $\mathfrak{d}_i = \left(\overline{\ell_{\mu^l(\mathfrak{S})}}(x^l), \overline{\ell_{\nu^l(\mathfrak{S})}}(x^l) \right)$ to construct a decision matrix $\overline{\mathfrak{R}}^i = \left[\mathfrak{Q}_{jk}^i \right]_{m \times n}$ for each $\mathcal{DM}_e \mathfrak{d}_i$.

Step 1: All attributes are considered benefit-type criteria. Accordingly, Eq. (12) is applied to obtain the normalized decision matrix for each decision maker.

$$\begin{aligned}\mathfrak{R}^1 &= [\mathfrak{Q}_{jk}^1]_{n \times n} = (\ell_{\mu^1(\mathfrak{S})}(x^i), \ell_{\nu^1(\mathfrak{S})}(x^i))_{3 \times 4} \\ \mathfrak{R}^2 &= [\mathfrak{Q}_{jk}^2]_{n \times n} = (\ell_{\mu^2(\mathfrak{S})}(x^i), \ell_{\nu^2(\mathfrak{S})}(x^i))_{3 \times 4} \\ \mathfrak{R}^3 &= [\mathfrak{Q}_{jk}^3]_{n \times n} = (\ell_{\mu^3(\mathfrak{S})}(x^i), \ell_{\nu^3(\mathfrak{S})}(x^i))_{3 \times 4}\end{aligned}$$

Step 2: Now, by using Eqs. (15) and (16) find the positive ideal alternative $(\mathcal{P}_{i\mathcal{A}})(\mathfrak{S}^+)^i$ and the negative ideal alternative $(\mathcal{N}_{i\mathcal{A}})(\mathfrak{S}^-)^i$.

Step 3: Eqs. (11), (15) and (16) obtain $(\mathfrak{C}_\omega^+)^i$ between $\mathfrak{S}_i$ and $(\mathcal{P}_{i\mathcal{A}})(\mathfrak{S}^+)^i$. Similarly, obtain $(\mathfrak{C}_\omega^-)^i$ between $\mathfrak{S}_i$ and $(\mathcal{N}_{i\mathcal{A}})(\mathfrak{S}^-)^i$ for each $\mathcal{DM}_e = \mathfrak{d}_i$.

For decision maker 1.

$$\begin{aligned}(\mathfrak{C}_\omega^+)^1 &= \mathfrak{C}_\omega(\mathfrak{S}_1, (\mathfrak{S}^+)^1) = 0.90, (\mathfrak{C}_\omega^-)^1 = \mathfrak{C}_\omega(\mathfrak{S}_1, (\mathfrak{S}^-)^1) = 0.38 \\ (\mathfrak{C}_\omega^+)^2 &= \mathfrak{C}_\omega(\mathfrak{S}_2, (\mathfrak{S}^+)^2) = 0.78, (\mathfrak{C}_\omega^-)^2 = \mathfrak{C}_\omega(\mathfrak{S}_2, (\mathfrak{S}^-)^2) = 0.50 \\ (\mathfrak{C}_\omega^+)^3 &= \mathfrak{C}_\omega(\mathfrak{S}_3, (\mathfrak{S}^+)^3) = 0.50, (\mathfrak{C}_\omega^-)^3 = \mathfrak{C}_\omega(\mathfrak{S}_3, (\mathfrak{S}^-)^3) = 0.84\end{aligned}$$

For decision maker 2.

$$\begin{aligned}(\mathfrak{C}_\omega^+)^1 &= \mathfrak{C}_\omega(\mathfrak{S}_1, (\mathfrak{S}^+)^1) = 0.87, (\mathfrak{C}_\omega^-)^1 = \mathfrak{C}_\omega(\mathfrak{S}_1, (\mathfrak{S}^-)^1) = 0.33 \\ (\mathfrak{C}_\omega^+)^2 &= \mathfrak{C}_\omega(\mathfrak{S}_2, (\mathfrak{S}^+)^2) = 0.75, (\mathfrak{C}_\omega^-)^2 = \mathfrak{C}_\omega(\mathfrak{S}_2, (\mathfrak{S}^-)^2) = 0.46 \\ (\mathfrak{C}_\omega^+)^3 &= \mathfrak{C}_\omega(\mathfrak{S}_3, (\mathfrak{S}^+)^3) = 0.46, (\mathfrak{C}_\omega^-)^3 = \mathfrak{C}_\omega(\mathfrak{S}_3, (\mathfrak{S}^-)^3) = 0.80\end{aligned}$$

For decision maker 3.

$$\begin{aligned}(\mathfrak{C}_\omega^+)^1 &= \mathfrak{C}_\omega(\mathfrak{S}_1, (\mathfrak{S}^+)^1) = 0.89, (\mathfrak{C}_\omega^-)^1 = \mathfrak{C}_\omega(\mathfrak{S}_1, (\mathfrak{S}^-)^1) = 0.34 \\ (\mathfrak{C}_\omega^+)^2 &= \mathfrak{C}_\omega(\mathfrak{S}_2, (\mathfrak{S}^+)^2) = 0.74, (\mathfrak{C}_\omega^-)^2 = \mathfrak{C}_\omega(\mathfrak{S}_2, (\mathfrak{S}^-)^2) = 0.47 \\ (\mathfrak{C}_\omega^+)^3 &= \mathfrak{C}_\omega(\mathfrak{S}_3, (\mathfrak{S}^+)^3) = 0.47, (\mathfrak{C}_\omega^-)^3 = \mathfrak{C}_\omega(\mathfrak{S}_3, (\mathfrak{S}^-)^3) = 0.83\end{aligned}$$

Step 4: Calculate aggregated $\mathfrak{C}_{\omega_i}^+$ and $\mathfrak{C}_{\omega_i}^-$ for the alternative $\mathfrak{J}_i$, using Eqs. (19) and (20).

$$\begin{aligned}\mathfrak{C}_{\omega_1}^+ &= 0.887, \mathfrak{C}_{\omega_1}^- = 0.353 \\ \mathfrak{C}_{\omega_2}^+ &= 0.760, \mathfrak{C}_{\omega_2}^- = 0.479 \\ \mathfrak{C}_{\omega_3}^+ &= 0.479, \mathfrak{C}_{\omega_3}^- = 0.824\end{aligned}$$

Step 5: Calculate the coefficient of closeness $\mathbb{C}_i$ using Eq. (21).

$$\begin{aligned}\mathbb{C}_1 &= \frac{0.887}{0.887 + 0.353} = 0.7153 \\ \mathbb{C}_2 &= \frac{0.760}{0.760 + 0.479} = 0.6134 \\ \mathbb{C}_3 &= \frac{0.479}{0.479 + 0.824} = 0.3676\end{aligned}$$

Step 6: Based on the closeness coefficients obtained using Eq. (21), the alternatives were ranked in descending order as $A_1 > A_2 > A_3$.

Accordingly, A_1 (Solar PV+) was identified as the preferred microgrid configuration, followed by A_2 (battery Solar PV + battery + backup diesel), while A_3 (diesel-only baseline) was ranked last. The result indicates that A_1 achieved the most favourable overall position relative to the positive ideal solution when the technical, economic, environmental, social, and resilience-related criteria were considered jointly.

6 Comparative Analysis and Theoretical Implications of the Proposed Model

The performance of the proposed framework was further examined through comparative and sensitivity analyses. The comparative analysis focused on whether the proposed correlation-based Lq-ROFHS framework produced preference rankings consistent with those obtained using existing MAGDM approaches. The sensitivity analysis was used to examine the stability of the decision outcomes under variations in the relevant model inputs and settings. Together, these analyses provide a basis for evaluating the behaviour of the proposed method and the stability of the resulting preference order. The comparative results are summarized in Table 2.

Table 2. Comparison of preference rankings obtained using the proposed and existing MAGDM approaches

MAGDM Method	Set Structure	Operators Used	Preference Order (Three Alternatives)	Preference Order (Four Alternatives)
MAGDM approach [33]	Lq-ROFs	Point aggregated and geometric operators prioritized weighted geometric aggregation operator	$A_3 < A_1 < A_2$	$A_3 < A_1 < A_4 < A_2$
MAGDM approach [34]	Lq-ROFs	Aggregated and weighted correlation coefficients	$A_3 < A_1 < A_2$	$A_3 < A_1 < A_4 < A_2$
Proposed method	Lq-ROFHSs	Aggregated and weighted correlation coefficients	$A_3 < A_2 < A_1$	$A_4 < A_3 < A_2 < A_1$

Note: MAGDM = multi-attribute group decision-making; Lq-ROFHSs = linguistic q-rung orthopair fuzzy hypersoft sets; Lq-ROFs = linguistic q-rung orthopair fuzzy sets.

Table 2 compares the preference orders obtained using the proposed method with those generated by the MAGDM approaches of Liu et al. [33] and Neelam et al. [34] under two comparative settings. For the three-alternative case considered in the microgrid application, all three approaches produced the same preference order, $A_3 < A_2 < A_1$ identifying A_1 as the preferred alternative. When a fourth alternative was included for further comparison, the three methods again produced an identical preference order, $A_3 < A_1 < A_4 < A_2$. These consistent ranking patterns indicate that the proposed method generates preference orders compatible with the two reference MAGDM approaches under both comparative settings.

The significance of the proposed framework, however, is not limited to reproducing the ranking obtained by existing approaches. Its methodological distinction lies in the integration of linguistic q-rung orthopair fuzzy information with a hypersoft structure and the proposed correlation measures. This formulation allows further subdivisions of decision attributes to be represented while retaining linguistic membership and non-membership information throughout the evaluation process. The resulting framework therefore provides an alternative analytical structure for decision problems in which conventional attribute representations may not adequately capture the granularity of the available information.

The comparison shown in Figure 3 indicates that the proposed method preserves the preference ordering obtained using the reference approaches. At the same time, the Lq-ROFHS formulation permits the evaluation to incorporate subdivided attributes together with linguistic membership and non-membership information. This feature is particularly relevant when the structure of a decision problem cannot be represented adequately by a single-level set of criteria.

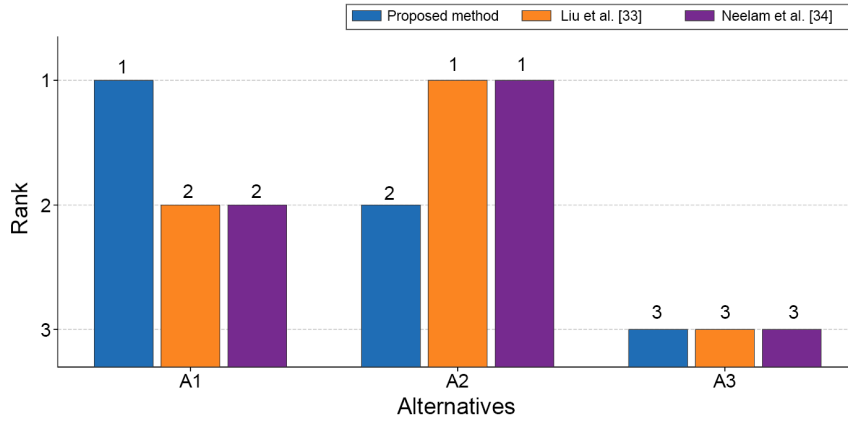


Figure 3. Comparison of preference rankings obtained using the proposed and existing multi-attribute group decision-making (MAGDM) approaches

6.1 Results Discussion and Managerial Implications

The numerical application demonstrates how the proposed framework can support resilient microgrid selection for rural healthcare facilities under multidimensional, uncertain, and linguistically expressed information. Although the present application focuses on microgrid selection, the analytical structure can also be adapted to other sustainability-oriented and smart-energy decisions in which precise numerical assessments are difficult to obtain. Potential applications include the selection of distributed energy resources, energy-storage technologies, renewable-energy configurations, and resilient grid strategies. The same decision structure may also be relevant to industrial and Industry 4.0 settings involving technology selection, supplier evaluation, manufacturing alternatives, and resource-allocation decisions based on heterogeneous expert assessments.

From a computational perspective, the proposed procedure consists of a sequence of structured operations involving Lq-ROFHS information representation, correlation calculation, weighting, aggregation, and TOPSIS-based ranking. These operations can be implemented in computational environments such as Python or MATLAB and may provide the analytical basis for an intelligent decision-support platform. The modular structure of the framework also allows additional alternatives, criteria, and decision makers to be incorporated without changing its fundamental decision logic. This characteristic is useful in sustainability-oriented decision settings in which the evaluation structure may evolve as new information, stakeholder assessments, or decision alternatives become available.

From a decision-support perspective, the framework provides a transparent mechanism for converting qualitative expert assessments into an interpretable preference order. This is particularly relevant to planning problems in which decision makers must balance several competing sustainability dimensions rather than optimize a single performance criterion. For resilient microgrid planning, the resulting ranking can support the systematic comparison of alternative configurations while retaining information related to technical performance, economic considerations, environmental effects, social suitability, and resilience. The model should therefore be viewed as a structured analytical aid to decision-making rather than as a substitute for contextual judgement by decision makers.

Several limitations should nevertheless be considered when interpreting the results. The reliability of the final ranking depends on the quality and consistency of the expert assessments, the linguistic quantifiers adopted in the evaluation, and the assumptions underlying the weighting, aggregation, and correlation procedures. Differences among expert judgements may influence the resulting preference order, particularly when the available information is highly uncertain. The computational requirements may also increase as the numbers of alternatives, criteria, sub-attributes, and decision makers grow. Further research is therefore needed to examine the scalability of the framework and to investigate its integration with optimization and other computational techniques for larger and more complex decision problems. Additional empirical applications would also be useful for evaluating the performance of the proposed approach across different sustainability and industrial decision contexts.

7 Conclusion

This study developed a sustainability-oriented decision-support framework by integrating Lq-ROFHSs with an extended TOPSIS approach. Correlation and weighted correlation coefficients were formulated to characterize relationships between Lq-ROFHS information, and their mathematical properties were examined. These measures were subsequently incorporated into a MAGDM procedure capable of representing linguistic uncertainty, hesitation, and further subdivisions of decision attributes within a unified analytical structure. The framework was applied to

resilient microgrid selection for a rural clinic, where alternative energy configurations were evaluated across cost efficiency, reliability, environmental benefit, and scalability criteria. Comparative and sensitivity analyses were also conducted to examine the behaviour of the proposed procedure and the stability of the resulting preference ranking.

The main contribution of the study lies in connecting correlation-based decision analysis with the richer information representation provided by the Lq-ROFHS structure. The proposed framework retains linguistic membership and non-membership information while allowing evaluation attributes to be decomposed into relevant sub-attributes. The comparison with existing MAGDM approaches produced a consistent preference order for the numerical example, indicating that the additional information structure can be incorporated without generating contradictory ranking behaviour. From a sustainability decision-analytics perspective, the framework provides a transparent and structured means of comparing alternatives when several competing dimensions must be considered simultaneously and expert assessments cannot be expressed reliably using precise numerical information.

The application to resilient microgrid selection illustrates the relevance of the proposed approach to sustainability-oriented energy planning. The framework can also be adapted to related decision problems, including smart-grid planning, renewable-energy integration, energy-storage selection, energy-management systems, and Industry 4.0 applications. Its modular structure allows additional alternatives, criteria, sub-attributes, and decision makers to be incorporated as the decision context evolves. Moreover, the computational procedure consists of structured operations that can be implemented in environments such as Python or MATLAB, providing a basis for its integration into intelligent decision-support systems.

Several limitations should be acknowledged. The resulting preference ranking depends on the quality and consistency of expert assessments, the specification of linguistic information, and the assumptions underlying the weighting, aggregation, and correlation procedures. The computational burden may also increase when substantially larger numbers of alternatives, criteria, sub-attributes, and decision makers are considered. Future research should therefore examine the scalability of the framework and validate its performance across additional empirical decision settings. Further extensions may integrate the proposed approach with optimization, machine learning, neural networks, Internet of Things technologies, and digital twins to support more dynamic and data-informed decision environments. Related decision-making applications in areas such as electric-vehicle infrastructure, general multiple-attribute selection, healthcare, finance, and environmental management [49, 51, 54] also provide directions for examining the broader applicability of the framework.

Author Contributions

Conceptualization, methodology, validation, formal analysis, writing—original draft preparation, and writing—review and editing: M.S.; Supervision, methodology, validation, formal analysis, project administration, and writing—review and editing: J.M.M. All authors have read and agreed to the published version of the manuscript.

Data availability

The data used to support the findings of this study are available from the corresponding author upon request.

Conflicts of Interest

The authors declare that they have no conflicts of interest.

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